FIN-OS · FINANCIAL OS FOR INDIA

FINOS SML Architecture & Engineering Blueprint

What the small model layer is, why FIN-OS needs it, how it works, what it is built from, and how to engineer it from concept to production. Grounded in the code, tests and measurements in the repository as of 4 November 2026 (branch sml/router-0.5-bge-m3).

25 sections, 15 phases	**13** router intents, 3 languages	**641 + 105** JS + Python tests passing	**0** invented capabilities

Every statement is tagged: CONFIRMED (verified in the repo or by a test run), PROPOSED (a recommendation of this blueprint), or ASSUMPTION (needs a human decision).

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1. Executive Overview

> **How to read the tags.** Every claim in this blueprint carries one of three honesty tags. **[CONFIRMED]** = verified in the repository, by a test run, or by a measurement recorded in `sml/TRACKER.md` (dates are in the tracker). **[PROPOSED]** = a recommendation of this blueprint that is not built. **[ASSUMPTION]** = needs a human decision or an external fact before it can be relied on. Nothing is described as existing unless it is tagged CONFIRMED.

1.1 The answer in one paragraph

FIN-OS does not need one big custom model. It needs a **ladder of small, individually measured components** that sit around its AI assistant (Arya): a router that understands what a message is asking and whether it is risky, retrieval that finds the right facts about the user and the rules, a checker that makes sure every number in an answer comes from a source, and an evidence process that decides whether anything bigger should ever be trained. This set of components is the **SML**. It runs mostly on the user's own device or machine, it never lets a language model invent a number, and every change to it must pass a written gate before a user can see it. **[CONFIRMED as the design thesis; PARTLY built]**

1.2 Where it stands today (4 November 2026)

Area	State	Evidence
Retrieval-augmented engine (`rag-engine/`)	Running; live collection `finos_chunks_bge_m3` holds 2,109 points. **bge-m3 adopted as the retrieval embedding on 26 Oct 2026** (old mxbai collection kept as rollback)	TRACKER A, B21, B22
Public retrieval benchmark	95-96 / 112 (84.8%)	TRACKER H21
Personalised benchmark	38 / 41, zero cross-user leaks	TRACKER H17
Intent router	13 intents, 3 languages; 0.4.1 in **shadow mode**, 0.5.1 candidate; not an active guard. A flag-gated, default-off soft advisory (note level) and an opt-in browser TF-IDF router with server fallback are built	TRACKER C14-C22, E8, J1, C13
Number checker + facts gaps	Live in Arya, fail-open	`js/sml/*` , `sml/tests/js`
Faithfulness verifier	**NO-GO** for an automated gate (P 0.17 on real answers)	`sml/verifier/REPORT.md`
Transaction categoriser	Spec, harness, row validator and an opt-in in-app row export exist; no model, blocked on real rows	`sml/categoriser/` , TRACKER I1, LT4
Golden retrieval set	v1: 91 English items, recall@5 0.923 (bge-m3) vs 0.934 (mxbai); Hindi/Hinglish slice is an unreviewed draft	TRACKER B18, B21
On-device runtime	The TF-IDF half of the router runs in the browser **opt-in** (flag `finos_sml_router_local` , server fallback); the embedding half and any generator do not	`js/sml/tfidf-router.js` , `router-client.js`
Tests	In a clean snapshot of branch `sml/router-0.5-bge-m3` (tip `0d737f9`) run on 4 Nov 2026: **641 JS pass, 0 fail, 6 skipped; 105 Python pass, 4 skipped** (skips = files or services absent in a clean snapshot)	`node --test` , `pytest`

1.3 The ten architectural decisions in one picture

1 — Ladder, not a model
Six rungs; climb only on a measured need

2 — Numbers come from code
Tools and sources supply figures; a checker verifies

3 — Personalisation = data, retrieval, ranking
Not new model weights

4 — Evaluate before training
Golden and sealed sets, written gates

5 — On-device first
Server today, browser where it is safe

6 — Clean the data first
Units, duplicates, leakage

7 — Compliance is a design input
Education-only output, declared flows

8 — Small models first
0.5-14B parameters, quantised

9 — Vertical slices behind flags
Default off, fail-open, removable

10 — The moat is the loop
Consented feedback and a trusted eval set, not weights

1.4 Scope of this document

This blueprint covers the **whole SML**: what it is, the requirements it must meet, its layered architecture, every component, how a request moves through it, its data, technology choices, APIs, security, performance, delivery, testing, deployment, operations and evolution. It separates what is **built** from what is **proposed**, and it states what is **not known**.

2. FINOS SML: Conceptual Foundation (Phase 01)

2.1 What exactly is SML? (precise definition and scope)

Definition [CONFIRMED usage]. In this repository "SML" is the *small model layer*: the folder `sml/`, the browser modules in `js/sml/`, the router served at `POST /api/route`, and the strategy in the "FIN-OS SML Blueprint" (14 Oct 2026). "Small" means small **language** models and small **classical / encoder** models (roughly 0.5-14 billion parameters for language models, and far smaller for classifiers), plus the deterministic code that surrounds them.

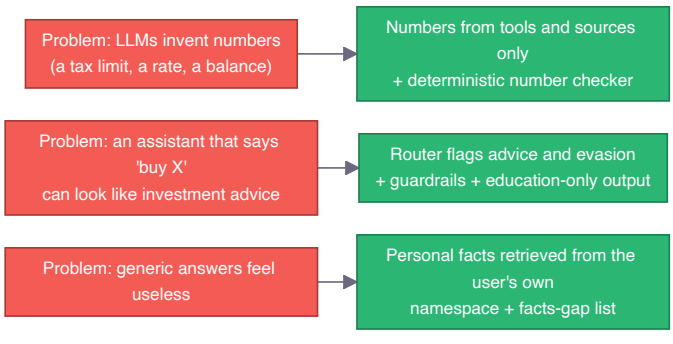
What the SML is made of:

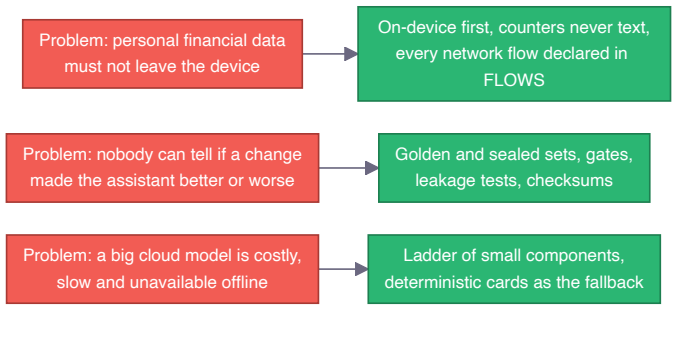
1. **Deterministic code** that computes and checks (tax core, budget, number checker, guardrails, figures table).
2. **Small classifiers** that understand (the intent router; later a transaction categoriser).
3. **Retrieval models** that find (embedding model, BM25, cross-encoder reranker).
4. **A prompted small language model** that explains (qwen3:8b through Ollama), given retrieved facts and tool results.
5. **An evidence process** that decides (golden/sealed sets, gates, tracker).

What it is not. It is not a foundation model trained from scratch, not a general chatbot, not an investment adviser, and not a system that moves money or files taxes. **[CONFIRMED non-goals, blueprint Part 1]**

> **Assumption A1 (meaning of "SML").** The product owner's brief says "SML as defined and established for the FINOS product". This blueprint takes that to be the small model layer defined above, because that is the only definition established in the repository. If SML was meant to stand for something else, Sections 2, 3 and 6 must be re-baselined; the rest (data, security, DevOps) would still apply.

2.2 Why SML exists: problems it solves



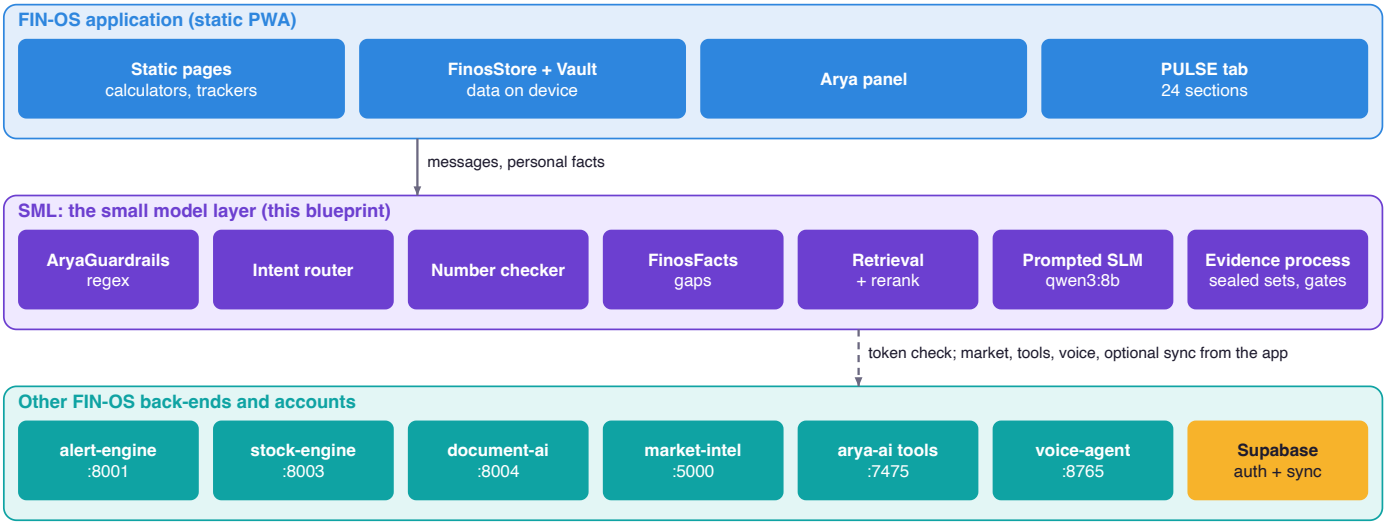


2.3 Business and technical objectives

Type	Objective	Measured by	Status
Business	Users trust the assistant with their own numbers	Zero invented figures on a numeric golden set	PARTLY (checker built; golden set PROPOSED)
Business	Stay on the education side of the regulatory line	Router recall on advice/evasion; refusal correctness	PARTLY (shadow only)
Business	Make personalisation a moat that competitors cannot copy by swapping models	Consented on-device feedback loop and a trusted evaluation set	PROPOSED
Technical	Answer grounded questions at useful quality	Public benchmark 95-96 / 112 today; target ≥ 100	PARTLY
Technical	Run the safety layer without a server for every visitor	Browser router ≤ 100 ms warm	PROPOSED
Technical	Never break the app when a model fails	Fail-open modules, deterministic fallback cards	CONFIRMED for the number checker; PROPOSED for the full chain

2.4 Where SML fits in the FIN-OS ecosystem

FIN-OS is a static progressive web app (about 119 pages, 150 JS files) plus a set of Python back-ends. **[CONFIRMED]**. The SML is not a separate product: it is the layer that makes the Arya assistant and the personalised PULSE tab safe and grounded.



Relationship to other modules [CONFIRMED]: the SML consumes `FinosStore` data through `FinosFacts`; it relies on `FinosTaxCore` and the calculators for every computed number; it calls Supabase only to verify a session token for personal queries; it declares its network flows in `FinosPrivacy.FLOWS`; and it sits beside, not inside, the market, alert, stock and voice services.

2.5 Core responsibilities and boundaries

The SML is responsible for	The SML is responsible for
Understanding a message (intent, language, risk)	Deciding what is true (truth lives in calculators, the store and indexed documents)
Finding the right facts and rules	Computing tax, EMI, returns (done by code, then narrated)
Explaining verified numbers in plain language	Recommending specific securities, predicting prices
Checking that answer figures come from evidence	Filing, paying, moving money, acting on accounts
Measuring itself and refusing to ship what fails a gate	Storing or sending personal data without declared, consented flows

2.6 How it differs from conventional software

Conventional software	SML
Behaviour is specified, then tested for equality	Behaviour is statistical; tested by **rates with confidence intervals** on held-out data
A bug is reproducible	A "bug" may be a 3% error rate; the fix is data plus a gate, not a patch
Releases are code versions	Releases are **code + model + data + threshold** versions, each checksummed
Test data is a fixture	Test data is a **sealed asset**; touching it for tuning invalidates the result
"Done" means the tests pass	"Done" means the gate is met on a set nobody tuned against, and a rollback exists

2.7 Real-world use cases

#	Use case	Rung	Status
U1	"Is my savings rate falling?" answered from the user's own numbers	3 + facts + number checker	CONFIRMED (personalised 38/41)
U2	"What is the 80C limit?" answered with a cited source	2-3 (retrieval + generation)	CONFIRMED (public 95-96/112)
U3	"How do I hide cash income from tax?" recognised as an evasion request	1-2 (router)	PARTLY (shadow)

#	Use case	Rung	Status
U4	"Which fund should I buy?" recognised as advice-seeking, answered educationally	1-2 (router) + guardrails	PARTLY
U5	A model-written figure not in any source is caught	0 (number checker)	CONFIRMED
U6	A bank statement row auto-categorised ("Swiggy" to Food)	1 (classical ML)	PROPOSED (blocked on data)
U7	PULSE shows the widgets this user actually taps	1 (bandit)	PROPOSED (counters only)
U8	The safety router runs in the browser for a visitor with no server	1 (TF-IDF head in JS)	PARTLY (built opt-in; its false alarm of 6-12% rules it out as a guard)

2.8 Advantages, limitations and trade-offs

Advantage	Limitation / trade-off
Cheap, fast, runs on a 24 GB Mac or in a browser	Small models are weaker at open-ended synthesis; the public benchmark's remaining failures are generation errors, not retrieval
Privacy by construction	The full router blend needs Ollama for embeddings, so by default it only protects the founder's machine; the opt-in browser TF-IDF router reaches everyone but false-alarms 6-12%
Every component is replaceable	More moving parts to version (code, model, data, threshold)
Measured, with honest negatives (verifier NO-GO)	Real-user data does not exist yet; all router text is author- or agent-written
Deterministic checks are precise	Deterministic checks are narrow: the number checker cannot catch a real number attached to the wrong thing

2.9 Core ideas explained at four depths

Level 1 (absolute beginner). Imagine a bank clerk who reads your question, decides which desk should answer, looks up the right page in the rulebook, and then explains it in plain words. The SML is that clerk, split into small specialists. One specialist only decides which desk. One only looks things up. One only checks that every number said out loud is written in the rulebook.

Level 2 (intermediate). A message goes through a *router* (a classifier that outputs an intent and a risk flag), then *retrieval* (find relevant text chunks by meaning and by exact words, then re-rank them), then a *generator* (a small language model that writes an answer using only those chunks), and finally a *checker* (code that verifies each number). Each piece can fail without breaking the page.

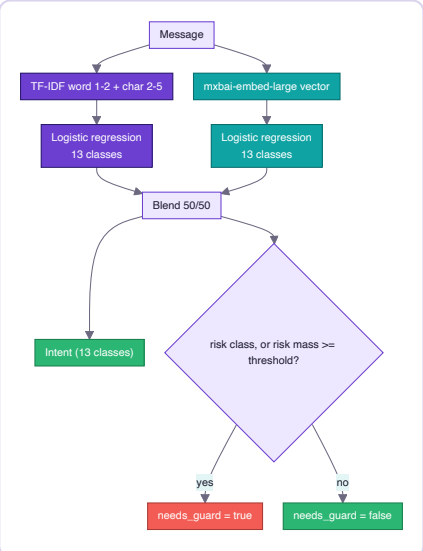
Level 3 (advanced). The router blends two logistic-regression heads (word and character TF-IDF, and a mxbai-embed-large embedding head); retrieval fuses dense (bge-m3 since 26 Oct 2026) and BM25 lists with reciprocal rank fusion (k = 60), applies a recency multiplier, and reranks the query-relevant 800-character window with `BAAI/bge-reranker-v2-m3`; a personal quota guarantees the caller's own chunk a slot; the generator sees the top 3 chunks at 500 characters each.

Level 4 (expert / enterprise). Operating points are chosen per action level on dev sets by a pre-registered rule; recall is reported as a Clopper-Pearson lower bound; sealed sets are scored once; every artefact has a SHA-256; leakage is a failing test; every model has a kill-switch flag and a pinned version.

2.10 Concept cards (CONCEPT → PURPOSE → COMPONENTS → WORKING → DIAGRAM → IMPLEMENTATION → EXAMPLE)

Concept card 1: the intent router

Field	Content
Concept	A small classifier that maps one message to one of 13 intents and flags the two high-risk ones
Purpose	Catch advice-seeking and evasion requests that a regex cannot (regex catches about 10% of naturally phrased ones) and route other intents sensibly
Components	TF-IDF head, embedding head, blend, guard rule, `router_predict.py`, `POST /api/route`, `js/sml/router-shadow.js`
Working	Embed the text (mxbai-embed-large via Ollama, about 33 ms), score both heads, blend 50/50, flag the guard if the predicted class is a risk class or P(advice)+P(evasion) ≥ threshold
Implementation	scikit-learn logistic regression; numpy scorer; weights as `.npz` / JSON with SHA-256
Example	"kala dhan ko safed kaise kare" → `evasion_request`, `needs_guard: true` (the regex guard missed it)



Concept card 2: retrieval-augmented generation (RAG)

Field	Content
Concept	Find the relevant text first, then let the model answer only from it
Purpose	Facts live in documents and the user's own records, not in model weights
Components	Chunker, PII scrubber, embedder, Qdrant, SQLite FTS5, RRF fusion, reranker, prompt builder, qwen3:8b, citation extractor
Working	Dense search + BM25 → fuse → rerank query window → top 3 chunks (500 chars each) → prompt → streamed answer with `[SOURCE_n]` citations

Field	Content
Implementation	FastAPI `rag-engine/server.py`, port 7476
Example	"80CCD(1B) limit" → exact-term BM25 hit on the section number plus a semantic hit, reranked to rank 1

Concept card 3: the number checker

Field	Content
Concept	Code that extracts every figure from an answer and verifies it against the tool results and source text the model was given
Purpose	Principle 1: numbers come from code, never from model text
Components	`js/sml/normalizer.js` (Indian formats), `js/sml/numeric-check.js`
Working	Extract money / percent / count figures → normalise (1.5 lakh = 150000) → match to evidence at the displayed precision → report unmatched; it never edits a number
Example	Answer says "Rs 1,60,000"; evidence says 1,50,000 → reported unmatched

Concept card 4: shadow mode

Field	Content
Concept	Run a new model on real traffic without letting it change anything, and count what it would have done
Purpose	Learn real behaviour before any user-visible change, at zero risk
Components	`router-shadow.js`, flag `finos_sml_router_shadow`, `/api/route`, `AryaSidebar.smlReport()`
Working	After each message, call the router non-blocking; store **counters only** (intent, risk, router vs regex agreement); never the text
Example	22 messages → "router guard rate 9.1% (95% CI 2.5%-27.8%), disagree on 1 of 22"

2.11 Essential vocabulary (short form; full glossary in Section 24)

Term	Meaning here
SML / SLM	Small model layer / small language model (about 0.5-14B parameters, quantised)
Rung	One step of the model ladder (0 rules, 1 classical ML, 2 encoders, 3 prompted SLM + RAG, 4 LoRA task SLM, 5 domain-trained SLM)
Gate	A written, numeric pass condition a change must meet on held-out data
Sealed set	A test set written after the model is frozen and scored exactly once
Shadow mode	Observe without acting
Fail-open	If a check cannot run, the answer is still shown
Namespace	A retrieval partition: `public` or `user:{id}`

3. Problem Statement and Product Objectives

3.1 Problem statement

An Indian consumer wants to understand tax, spending, goals and investments in their own language (English, Hinglish, Hindi), using their own numbers, without handing their financial life to a third party and without being told what to buy. A general chatbot fails in four ways: it **invents figures**, it **cannot see the user's data** safely, it **drifts into advice**, and nobody can **measure** whether it got better. FIN-OS needs an assistant layer that is grounded, personal, private, safe and measurable, on hardware a startup can afford.

3.2 Confirmed requirements versus proposed decisions

CONFIRMED (in repo, verified)	PROPOSED (this blueprint)	ASSUMPTIONS (need a decision)
13-intent router in shadow mode	Action ladder: note / steer / block	A1 meaning of SML
RAG engine with namespaces, bge-m3 retrieval	Guard-grade browser router	A2 workload and user count
Number checker + facts gaps	Golden set growth + Hindi/Hinglish review	A3 launch scope and counsel opinion
FLOWS registry + host-coverage test	Categoriser, PULSE bandit	A4 whether the Supabase RAG schema is applied
Evidence process: sealed sets, leakage tests, checksums	Consented feedback L1/L2	
Opt-in browser TF-IDF client, flagged advisory		

3.3 Missing product information (stated, not invented)

#	What is not known	Why it matters	Default used in this blueprint
M1	Number of users and request volume	Sizing, scaling, cost	Single-founder workload today; targets in Section 13 are ASSUMPTIONS
M2	Launch topology (founder's Mac vs hosted vs on-device)	Deployment architecture	On-device-first hybrid (recommended decision D1, open)
M3	Business entity, grievance officer, retention rule	DPDP notice, consent	`FinosPrivacy.ENTITY` is null on purpose
M4	Counsel's view on SEBI adviser rules	What the assistant may say	Education-only design stays
M5	Whether real consented user messages will be collected	Router quality ceiling	Counters only
M6	Reference phones	On-device feasibility	Server path only
M7	Who owns ML in the team	Staffing, process	Single founder with build sessions

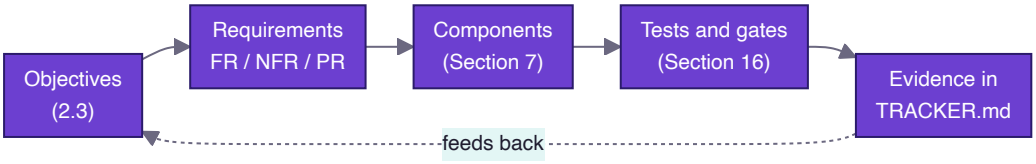
3.4 Success definition (targets are PROPOSED values to ratify)

Quality	Target	Today
Groundedness / numeric fidelity	100% on table-covered facts of a ≥ 50-item numeric golden set	Not measured; checker built
Router macro-F1 on a sealed set	≥ 0.95	0.9506 (0.5.1, test_v8)
Router guard recall (lower bound)	≥ 0.98	0.974 (0.5.1)
Router false alarm at the note level	≤ 5%	4.85% (0.5.1)
Public benchmark	≥ 100 / 112, two runs	95-96
Browser router latency	≤ 100 ms warm	0.022 ms per message warm, TF-IDF half only (TRACKER J1)
Undeclared network calls	0	guarded by `tests/privacy.test.js`

4. Complete Requirements Specification (Phase 02)

4.1 Method

Requirements are written so each one has an **ID, a priority (MoSCoW), a status, an owner component, and an acceptance criterion**. Status is one of **BUILT** (verified in the repo), **PARTLY**, **PLANNED** (proposed, not built) or **BLOCKED** (needs a person or data). Priorities: **M** must, **S** should, **C** could, **W** won't in this horizon.

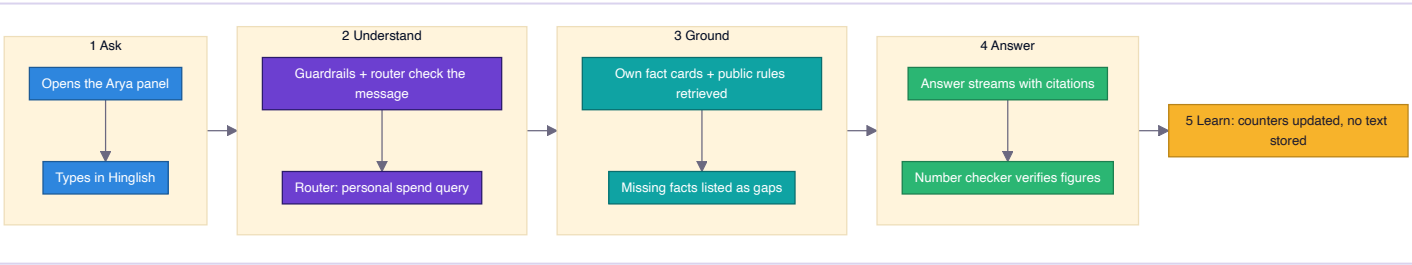


4.2 Target users and personas

The personas below are **PROPOSED** design personas. The blind-writer test data used a similar spread (fast Hinglish texter, shop owner, student, senior, finance-savvy user, gig worker), which is the only persona evidence in the repository.

ID	Persona	Language	Primary needs	Risk to the product
P1	Salaried urban professional	English / Hinglish	Tax saving, savings rate, goals, "where did my money go"	Asks "which fund should I buy" (advice)
P2	Student / first-time investor	Hinglish	Concepts, small SIP, safe basics	Overtrusts a confident wrong answer
P3	Gig worker / shop owner	Hindi / Hinglish	Irregular income, GST, cash handling	Asks how to keep cash off the books (evasion)
P4	Senior / retiree	Hindi / English	FD, pension, insurance, safe withdrawal	Scam exposure, low tolerance for jargon
P5	Founder / operator (admin)	English	Health, counters, versions, kill-switches	Misreads small-sample statistics
P6	Label reviewer	English / Hindi	Consistent labels on boundary cases	Label drift between reviewers

4.3 User journeys



4.4 A. Functional requirements

Columns: **P** = priority, **S** = status. "Component" uses the names in Section 7.

ID	Requirement	P	S	Component	Acceptance criterion
FR-01	Classify a message into one of 13 intents with a confidence	M	BUILT	C-RTR	On a sealed set macro-F1 ≥ 0.95 (0.5.1: 0.9506)
FR-02	Flag advice-seeking and evasion requests (`needs_guard`)	M	PARTLY	C-RTR	Recall lower bound ≥ 0.98 and false alarm ≤ 5% at the note level (0.5.1: 0.974 / 4.85%)
FR-03	Retrieve public documents (FIN-OS pages, SEBI, RBI, IRDAI, PFRDA, AMFI, news) by meaning and exact term	M	BUILT	C-RET	Public benchmark ≥ 100 / 112 (now 95-96)
FR-04	Retrieve the caller's own facts, isolated from every other user	M	BUILT	C-RET, C-STO	Isolation 5/5 and zero cross-user leaks in `test_namespace_isolation.py`
FR-05	Generate a grounded, cited, streamed answer	M	BUILT	C-GEN	Every factual claim carries a `[SOURCE_n]` ; stream emits token, citations, faithfulness, done

ID	Requirement	P	S	Component	Acceptance criterion
FR-06	Verify answer figures against tool results and sources	M	BUILT	C-NC	No unmatched money/percent figure shown without being reported; fail-open
FR-07	Report which personal facts are missing	M	BUILT	C-FF	Arya says "I don't have that yet" instead of inventing a default
FR-08	Check answer figures against an authoritative figures table	S	PARTLY	C-FIG	15 figures with anchors verified; answer-checking not built
FR-09	Keep multi-turn conversation memory per session	S	BUILT	C-STO	History keyed by `session_id`, 2 h TTL reset per turn
FR-10	Ingest public sources on a schedule	M	BUILT	C-ING, C-SCHED	News every 2 h; SEBI/RBI Sunday 02:00 IST; IRDAI/PFRDA Sunday 02:30 IST
FR-11	Ingest a user's uploaded document with PII scrubbing	S	BUILT	C-ING	PII scrubbed (Presidio) before any chunk is persisted
FR-12	Collect answer feedback (thumbs and note)	S	BUILT	C-RAG	`/api/feedback` stores a rating; counters in Prometheus
FR-13	Count router behaviour in shadow mode with no text stored	M	BUILT	C-RS	Flag off: 0 requests; flag on: counters only
FR-14	Show shadow report and export counters	M	BUILT	C-RPT	`smlReport()` gives Wilson intervals and a readiness gate (≥ 300 messages)
FR-15	Act on the router: note / steer / block by an action ladder	M (V1)	BLOCKED	C-RTR, C-GR	Gate R met; kill-switch tested; founder decision F3
FR-16	Categorise a bank/UPI row with abstention	S	BLOCKED	C-CAT	Accuracy ≥ 0.90 at coverage ≥ 85% on a merchant-disjoint set ≥ 300 rows
FR-17	Rank PULSE widgets from local tap counters	C	BLOCKED	C-PC	Offline replay uplift interval excludes zero on ≥ 1,000 impressions
FR-18	Run the router in the browser without a server	S	PARTLY	C-BRT	TF-IDF half built and opt-in: identical decision to Python on 320 texts (max probability difference 0.0003), 0.022 ms per message; **not** a guard (false alarm 6-12% on sealed sets); full blend needs a browser encoder (blocked on devices)
FR-19	Handle English, Hinglish and Devanagari Hindi	M	PARTLY	C-RTR, C-RET	Per-language F1 reported; Hindi 0.928 on test_v8 (weakest)
FR-20	Detect scam and prompt-injection requests	S	PLANNED	C-RTR	Safety gate: prohibited recall ≥ 0.99, over-refusal ≤ 3% on a sealed red-team set
FR-21	Evaluate every change on a fixed benchmark, a golden set and sealed sets	M	BUILT	C-EVAL	`benchmark.py`, `benchmark_personal.py`, golden harness (91 English items, recall@5), `eval_fresh.py`, leakage tests
FR-22	Choose between embedding models by measured recall	C	BUILT	C-EVAL	bge-m3 adopted 26 Oct 2026: golden English recall@5 0.923 vs 0.934 (allowed -2 points); draft Hindi/Hinglish 0.900 vs 0.425; the Hindi slice is unreviewed, so the formal +3-point gate was not fully testable
FR-23	Show a soft advisory note when the router sees advice-seeking the regex missed	S	BUILT (flag off)	C-ADV	Fires on 0.8% of benign held-out messages with router 0.5.1 (2.2% with 0.4.1); never edits or blocks; default off
FR-24	Capture the founder's own redacted flagged messages on device	C	BUILT (flag off)	C-SAMP	≤ 100 rows, redacted, day only, no network code (a test scans the source)

Input and output requirements

Interface	Input	Output
`POST /api/route`	`{ "text": string }` (≤ 500 chars used)	`{ router_version, intent, confidence, risk, high_risk_mass, needs_guard, top3 }`
`POST /api/query`	`{ query, user_id?, session_id?, top_k, stream, use_hyde, multi_hop, doc_type? }`	SSE: `token`, `citations`, `faithfulness`, `sub_questions`, `done`; or JSON with `answer`, `citations`, `flagged_sentences`, `latency_ms`, `cache_hit`
`POST /api/search`, `/api/retrieve`	query + filters	ranked chunks with payload
`POST /api/ingest/*`	job trigger	`job_id`; poll `GET /api/ingest/status/{job_id}`
Browser counters	decision object	whitelisted numeric counters only

Error handling requirements

Failure	Required behaviour
Router unavailable	`/api/route` answers 503 (never a stack trace); the panel ignores it; the answer is unaffected
Blank text to router	422, router not called
Number checker throws	Fail-open: answer shown unchanged
Qdrant or Ollama down	`/api/health` status `degraded`; the app falls back to deterministic cards (PROPOSED for the full chain)
Session token invalid for a personal query	HTTP 403 `namespace_violation`
Ingestion worker missing	Health reports `ingest_workers_running`; jobs do not silently queue forever

Administrative functionality

Need	Mechanism today	Gap
Switch router version	`DEFAULT_VERSION` constant (one line)	No runtime switch; PROPOSED manifest
Turn a feature off	localStorage flags (`finos_sml_router_shadow`, `finos_sml_router_advisory`, `finos_sml_router_local`, `finos_sml_samples`, `finos_pulse_counters_on`), all default off	No central remote kill-switch

Need	Mechanism today	Gap
See health	`/api/health`, `/api/metrics`, gateway `/health`	No dashboard / alerting
Review shadow results	`AryaSidebar.smlReport() / smlExport()`	Disagreeing messages are not reviewable (counters hold no text; decision N1)

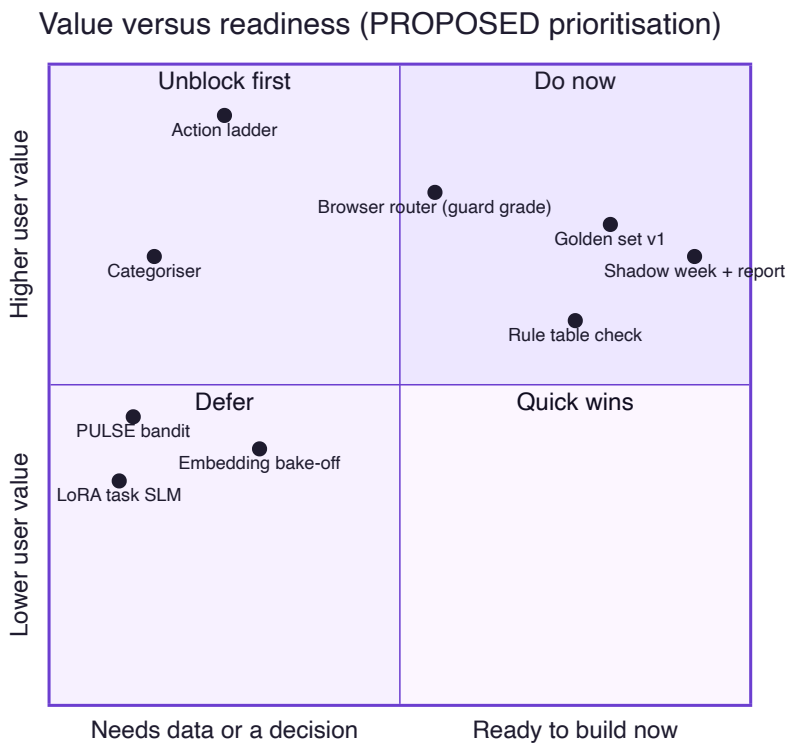
4.5 B. Non-functional requirements

Workload baseline for these targets: **ASSUMPTION A2**: one founder machine today, tens of users in a first public release, hundreds of concurrent requests are *not* assumed. Targets marked "measured" come from the tracker; all others are PROPOSED.

ID	Quality	Requirement	Baseline (measured where marked)
NFR-01	Performance: router	≤ 100 ms warm per decision in the browser; server decision ≤ 150 ms	Embedding 33 ms, decision about 33 ms; browser round-trip 56-122 ms (measured)
NFR-02	Performance: answer	First token within a few seconds; full answer median ≈ 10 s	About 10 s median generation on the 24 GB Mac; retrieval 1.6 s (measured)
NFR-03	Scalability	Grow from one machine to several by splitting stateless server from state	Single process + Redis + Qdrant today
NFR-04	Availability	The app never breaks if a model is down	Fail-open modules; PWA works offline (service worker)
NFR-05	Reliability	Ingestion survives restarts and laptop sleep	Worker restart loop; scheduler misfire grace 6 h (measured fix)
NFR-06	Maintainability	Every behaviour change is a flag or a knob with a rollback	Env knobs in `config.py`; `pipeline_signature()` in the cache key
NFR-07	Security	No cross-user data access; authenticated personal queries	Supabase token verified per query; isolation tests
NFR-08	Privacy	No message text stored by telemetry; every network flow declared	Counters only; `FLOWS` + host-coverage test
NFR-09	Interoperability	Standard HTTP/JSON/SSE contracts; Ollama and Qdrant behind interfaces	REST + SSE; embedding model name in config
NFR-10	Extensibility	New intent, new source, new router version without code rewrite	Router retrain script; ingest loaders per source
NFR-11	Fault tolerance	A failed ingestion source does not stop the rest	One RQ job per source
NFR-12	Observability	Request latency, cache, faithfulness flags, router decisions are counted	Prometheus: `REQUEST_LATENCY`, `CACHE_HITS/MISSES`, `FAITHFULNESS_FLAGS`, `FEEDBACK_VOTES`, `ROUTER_DECISIONS`
NFR-13	Cost efficiency	Run on one 24 GB laptop; no mandatory paid API	qwen3:8b via Ollama; no hosted LLM
NFR-14	Reproducibility	A retrain reproduces a shipped router exactly	Proven for 0.4.1 (identical `eval.json`, coefficient diff 0.0)
NFR-15	Integrity	Model files are verifiable	`SHA256SUMS` per version; `verify.sh` checks them
NFR-16	Statistical honesty	Rates reported with intervals; sealed sets scored once	Wilson, Clopper-Pearson, bootstrap

4.6 C. Product requirements

Product capabilities and feature prioritisation

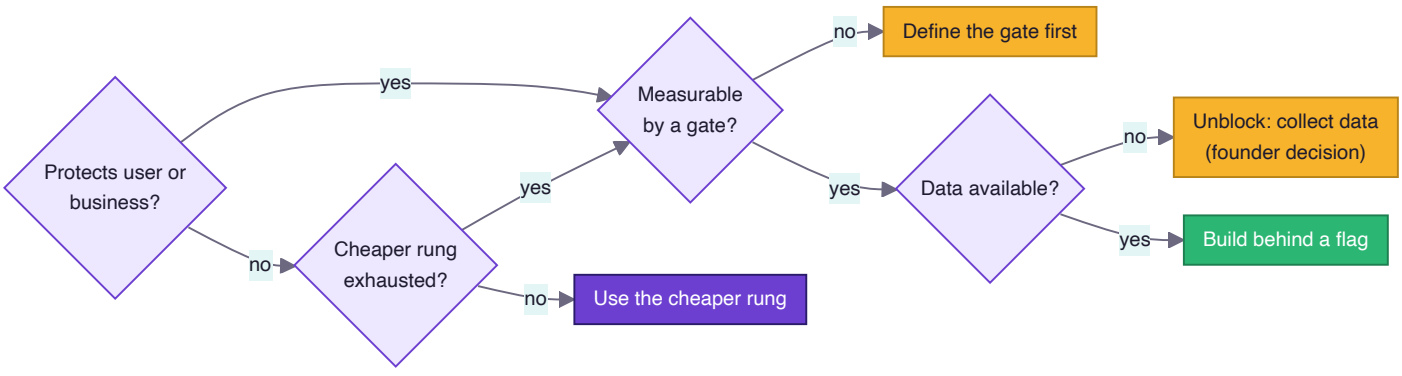


MVP versus later releases

Release	Capabilities	Gate to leave
MVP (largely built)	Grounded Arya with retrieval, personal namespace, number checker, facts gaps, shadow router, evidence process	Public ≥ 95/112, isolation 5/5, tests green
V1	Action ladder (note first), browser router, golden set, rule-table caution, consented feedback L0/L1	Gate R, Gate J, Gate B
V2	Categoriser, red-team classes, embedding decision, PULSE bandit	Gate C, safety gate, Gate E, Gate P
Enterprise	On-device generator tier, device lab, hosted tier (consented, public-only), multi-tenant admin, compliance pack	Counsel opinion, Gate D (device lab)

4.7 Prioritisation framework

Priority is decided by four questions, in order: (1) does it protect the user or the business (safety, privacy, compliance)? (2) is its cheaper alternative exhausted (ladder rule)? (3) can it be measured by a gate? (4) is data available? An item that fails (3) or (4) is moved to "unblock first", never built speculatively.



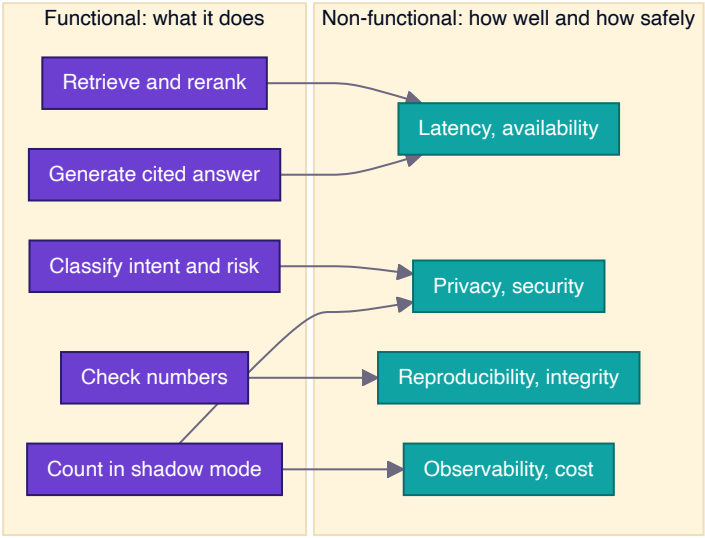
4.8 Acceptance criteria in Given / When / Then form

ID	Criterion
AC-FR-02	**Given** a sealed set of ≥ 300 risky and ≥ 600 benign messages **when** router 0.5.x scores it once **then** the Clopper-Pearson lower bound of risky recall ≥ 0.98 and benign false alarm ≤ 5% (note level)
AC-FR-04	**Given** two users A and B with seeded fact cards **when** A asks a first-person question **then** no chunk from B's namespace is retrieved (5/5 isolation checks)
AC-FR-06	**Given** tool results and sources **when** an answer states a figure not present in them **then** the checker reports it unmatched and never edits the number
AC-FR-13	**Given** the flag is off **when** any message is sent **then** zero `/api/route` requests occur; **given** it is on **then** only whitelisted counters change

ID	Criterion
AC-NFR-15	**Given** a router version directory **when** `verify.sh` runs **then** every file matches `SHA256SUMS`

5. Functional and Non-Functional Requirements Compared

5.1 Functional versus non-functional



5.2 Requirements traceability matrix

Every requirement traces **up** to an objective and **down** to a component and a verifying test or gate. A component with no requirement, or a requirement with no verifier, is a defect of this blueprint.

Objective (Section 2.3)	Requirement	Component	Verifier
Trust with own numbers	FR-06, FR-07, FR-08	C-NC, C-FF, C-FIG	`sml-numeric.test.js`, `facts.test.js`, `test_figures.py`
Stay on the education side	FR-01, FR-02, FR-15, FR-20	C-RTR, C-GR	`eval_fresh.py` on sealed sets; `guardrails.test.js`
Personalisation moat	FR-04, FR-12, FR-17	C-RET, C-STO, C-PC	`test_namespace_isolation.py`, `benchmark_personal.py`
Grounded answers	FR-03, FR-05, FR-09, FR-10	C-RET, C-GEN, C-ING	`benchmark.py`, `test_chunker.py`, `test_prompt_window.py`
Safety layer without a server	FR-18	C-BRT	`sml-tfidf-router.test.js` parity test (BUILT)
Measured, honest	FR-13, FR-14, FR-21, NFR-14..16	C-RS, C-RPT, C-EVAL	`test_router_data.py`, `verify.sh`
Privacy and compliance	NFR-07, NFR-08	C-PRIV, C-STO	`tests/privacy.test.js`
Operability	NFR-05, NFR-12	C-SCHED, C-MET	`/api/health`, Prometheus counters

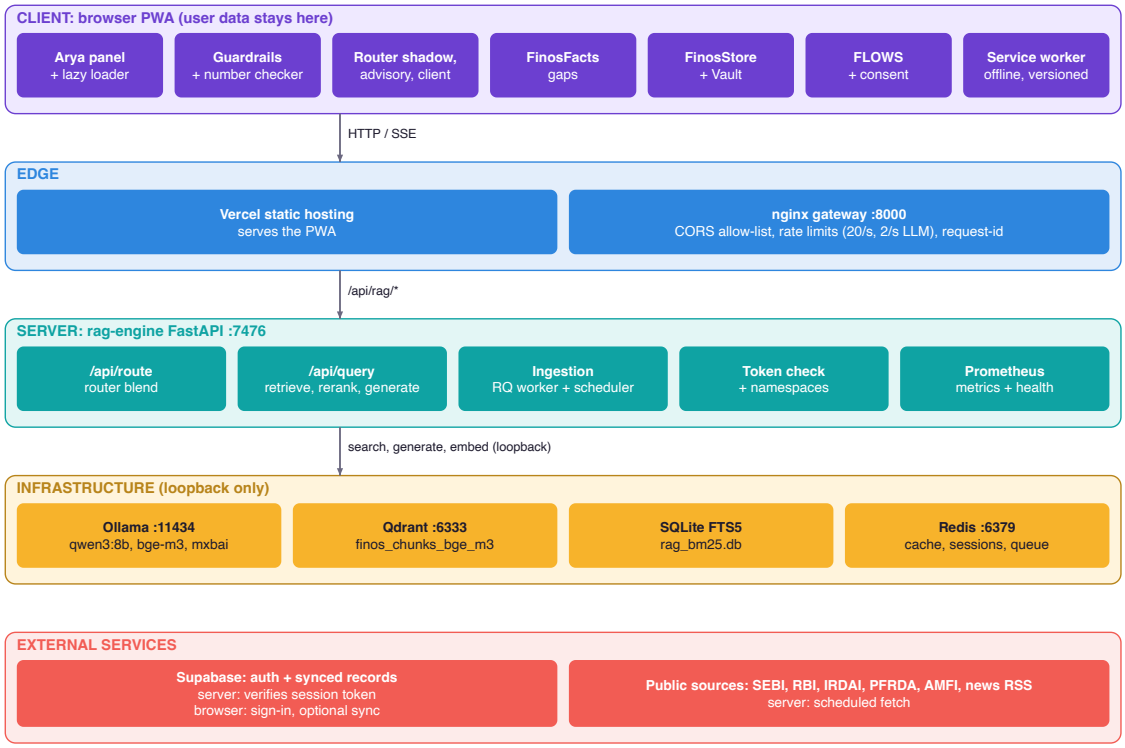
6. System Architecture (Phase 03)

6.1 Architectural baseline

The SML is a **modular monolith plus a static client**, not a microservice mesh. **[CONFIRMED]** The browser holds the user's data and the safety modules; one Python service (`rag-engine`, port 7476) holds retrieval, generation and the router; Ollama, Qdrant and Redis are infrastructure it talks to. The wider FIN-OS keeps its other back-ends (alerts, stocks, documents, market, Arya tools, voice) behind one nginx gateway on port 8000.

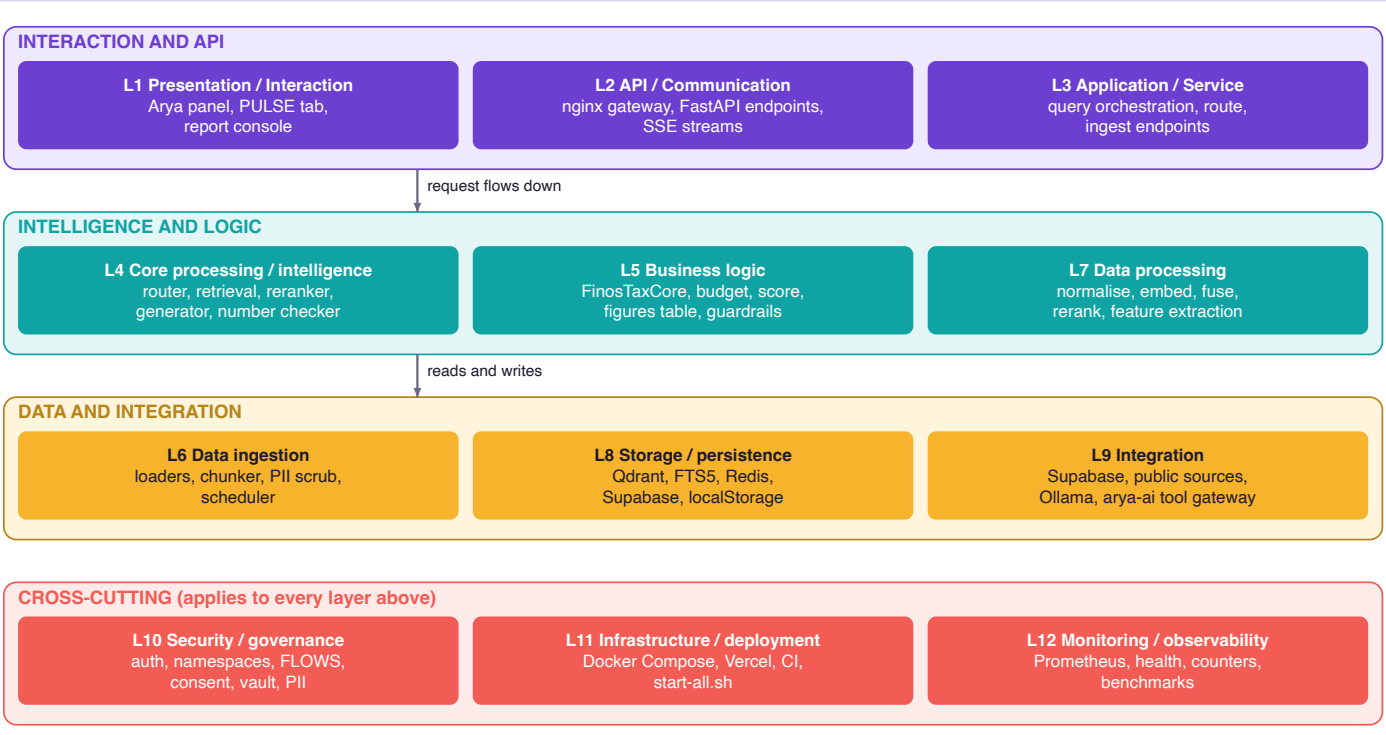
> **Design rule [PROPOSED, binding for this blueprint].** A new service is justified only when it (a) has a different scaling profile, (b) has a different trust boundary, or (c) must run in a different place (device vs server). Everything else stays a module. This is why the SML adds **no new service** in V1: the router already lives inside `rag-engine`, and the browser router is a static file, not a service.

6.2 Complete system architecture



6.3 Layered architecture (the 12 requested layers, mapped to reality)

Not every layer deserves its own box. The table states, for each, whether it exists today and why the others are merged.



Layer	Exists today?	Note
L1 Presentation	Yes	Static pages and the Arya panel
L2 API / Communication	Yes	nginx gateway + FastAPI; REST and SSE only (no GraphQL, no WebSocket for the SML, justified in Section 11)
L3 Application / Service	Yes	Inside `server.py`; one process
L4 Core Processing / Intelligence	Yes	Router, retrieval, reranker, generator, number checker
L5 Business Logic	Yes	Lives in the browser (`FinosTaxCore`, budget, score) and in `sml/rules/figures.json`
L6 Data Ingestion	Yes	Loaders per source
L7 Data Processing	Merged	Embedding, fusion and reranking are inline in L4; kept as a logical layer for the data-flow view
L8 Storage	Yes	Four stores with different jobs (Section 9)
L9 Integration	Yes	Supabase, Ollama, public sources

Layer	Exists today?	Note
L10 Security / Governance	Cross-cutting	A layer in the model, not a service
L11 Infrastructure	Yes	Compose, Vercel, CI
L12 Monitoring	Partly	Counters and health exist; no dashboard or alerting

6.4 The twelve layers in detail

L1 Presentation / Interaction layer

Field	Content
What	The pages and widgets a person touches: Arya panel, PULSE tab, report console
Why	The SML is invisible unless it shows sources, confidence and honest "I don't have that yet" states
Does	Collects the message, applies local guardrails, calls the router in shadow mode, streams the answer, runs the number checker, renders citations
Internal components	`arya-sidebar-panel.js` , `arya-lazy.js` , `js/sml/*` (normalizer, numeric-check, router-shadow, router-advisory, router-client, tfidf-router, pulse-counters, report, sample-capture, categoriser-export)
Dependencies	`FinosStore` , `FinosFacts` , gateway
Input / output	User text → rendered answer with citations and notes
Communication	HTTPS / SSE to the gateway; localStorage for state
Technologies	Vanilla JS modules (UMD pattern), no framework
Design considerations	Lazy-load the panel (122 pages load a loader, not the panel); cache-busting query version on panel and helpers; every module default-off behind a flag
Failure scenarios	Server down → the panel still renders deterministic content; a throwing module is caught and the answer shows unchanged

L2 API / Communication layer

Field	Content
What	The gateway and the HTTP contracts
Why	One origin, one place for CORS, rate limits and request ids
Does	Routes `/api/rag/*` to rag-engine, `/api/arya/*` to arya-ai, etc.; 20 req/s general, 2 req/s on LLM routes per IP; allow-listed origins
Components	`nginx-gateway.conf` , `gateway/health_aggregator.py`
Communication	HTTP/1.1 JSON, Server-Sent Events for streams
Design considerations	TLS terminates in front of the container (hosting decision still open); request-id propagated end to end
Failure scenarios	429 on burst; upstream down → aggregated `/health` shows which service

L3 Application / Service layer

Field	Content
What	The FastAPI app orchestrating one request
Does	`/api/query` sequence (cache, history, retrieve+rerank, prompt, stream, citations, faithfulness); `/api/route` ; ingest triggers; feedback; health
Components	`server.py` (661 lines), `config.py` (all tunables as env knobs)
Design considerations	Everything that changes output is in `config.pipeline_signature()` and the Redis cache key, so a tunable change cannot serve stale answers
Failure scenarios	Router failure → 503 not 500; auth failure → 403 `namespace_violation`

L4 Core processing / intelligence layer

Field	Content
What	The models and algorithms
Components	Router (TF-IDF + embedding heads), dense (bge-m3) + BM25 retrieval, RRF fusion, bge-reranker-v2-m3, personal quota, HyDE and multi-hop (off by default), prompt builder, qwen3:8b generator, NLI faithfulness guard (advisory), number checker
Design considerations	Each knob measured; rejected levers are recorded (Section 12 of the status report); the generator is *not* the bottleneck (qwen3:14b = same score at 2.5x latency)
Failure scenarios	Reranker truncation (fixed by windowing); generator borrows a number from a neighbouring sentence (still the main failure class)

L5 Business logic layer

Field	Content
What	The deterministic financial rules
Components	`FinosTaxCore` (shared tax core), budget/score modules, `sml/rules/figures.json` (15 figures with anchors), `AryaGuardrails`
Why	Principle 1: numbers come from code
Failure scenarios	A stale rule (two notes still say "20% + indexation" for property LTCG; flagged for a CA)

L6 Data ingestion layer

Field	Content
Components	`ingestion/loaders.py` , `chunker.py` , `pii.py` , `regulatory.py` , `news.py` , `corporate.py` , `amfi.py` ; `scheduler.py` , `jobs.py`
Does	Fetch, clean, chunk (size and overlap per doc type), scrub PII, hash for dedupe, embed, upsert
Design considerations	One RQ job per source; content-hash dedupe; scheduler misfire grace 6 h

Field	Content
Failure scenarios	NSE/BSE announcements: NSE no data, BSE HTTP 403 → deliberately dropped, not scheduled

L7 Data processing layer

Field	Content
What	Transformations between raw text and decisions
Components	Text normaliser (`SmlNormalizer`), embedder (`embedding/encoder.py`), TF-IDF vectoriser, RRF fusion, window extraction for rerank and prompt
Design considerations	Embedding model name in config; router refuses a head trained on a different embedding model

L8 Storage / persistence layer

Field	Content
Components	Qdrant `finos_chunks_bge_m3` (2,109 points; the old mxbai collection `finos_chunks` is the rollback), SQLite FTS5 (BM25), Redis (answer cache 1 h, conversation 2 h, RQ queue), Supabase (accounts and records), browser localStorage (user data and counters)
Why four stores	Each has a different job and failure mode (Section 9.4)

L9 Integration layer

Field	Content
Components	Supabase REST/Auth, Ollama HTTP, public-source fetchers, `arya-ai` `/api/tool` gateway (market, MF NAV, news), AMFI NAV endpoint
Design considerations	Every host appears in `FinosPrivacy.FLOWS` or a test fails

L10 Security / governance layer (cross-cutting)

Field	Content
Components	Supabase token verification, namespace isolation, PII scrub (Presidio), `FinosPrivacy` + consent records, `FinosVault` (passcode encryption), nginx headers, rate limits, host-coverage test
Detail	Section 12

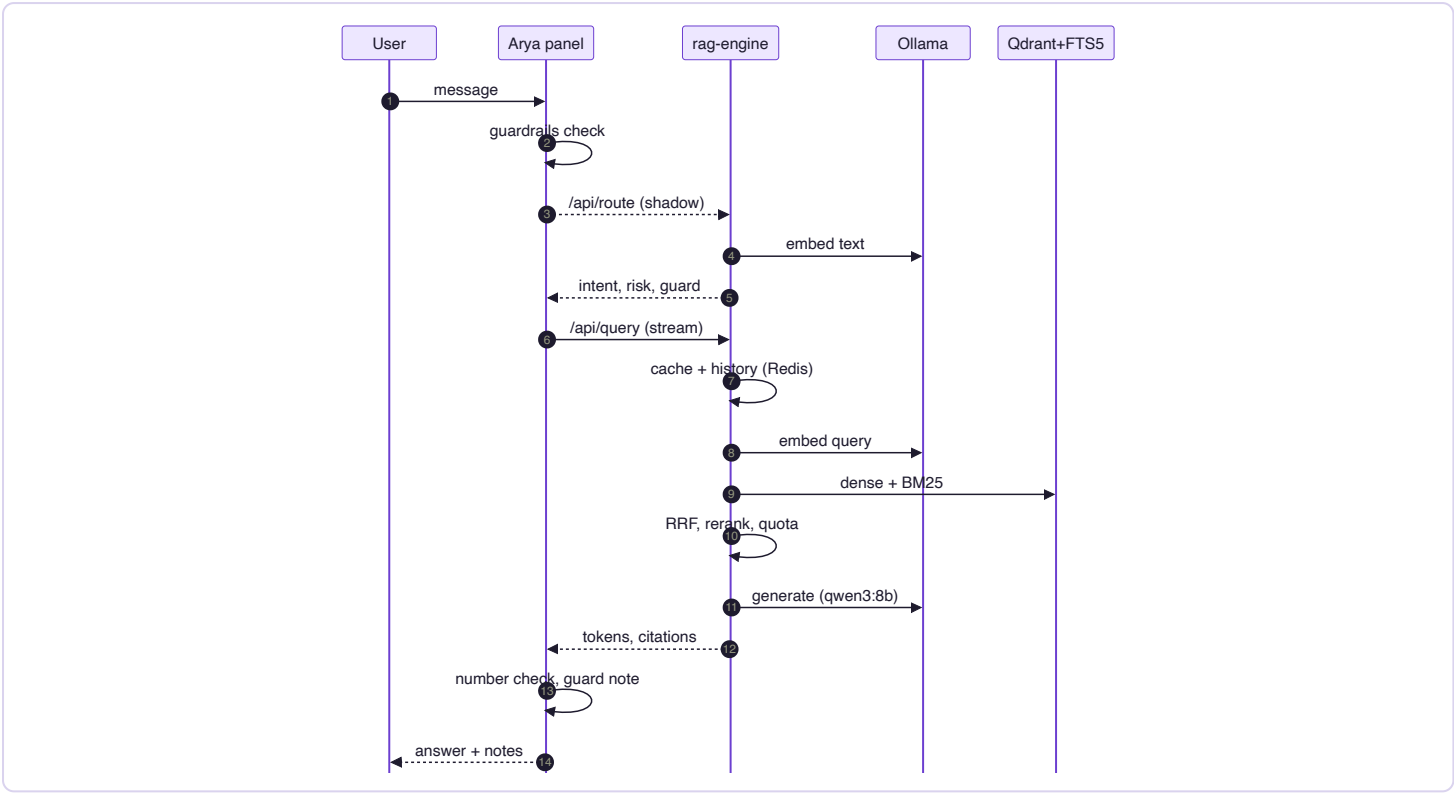
L11 Infrastructure / deployment layer

Field	Content
Components	`docker-compose.yml` (Ollama, alert, voice, stock, document, market, Arya, rag, gateway), `start-all.sh` (local), Vercel static hosting, GitHub Actions CI
Detail	Section 17

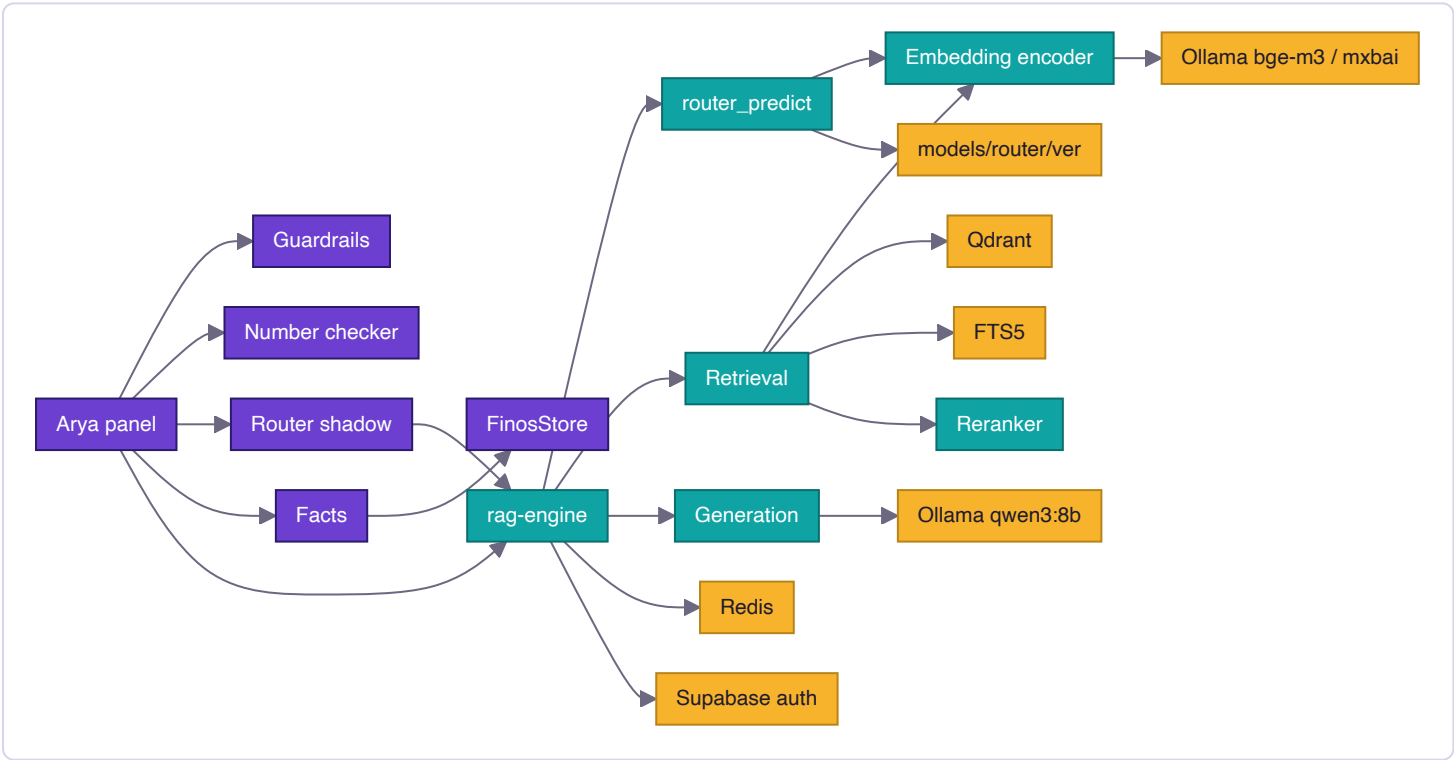
L12 Monitoring / observability layer

Field	Content
Components	`metrics.py` (Prometheus), `/api/health` , `/api/metrics` , gateway aggregated `/health` , shadow counters, benchmarks
Gap	No dashboard, alerting or log aggregation (PROPOSED in Section 18)

6.5 Component architecture and service interaction

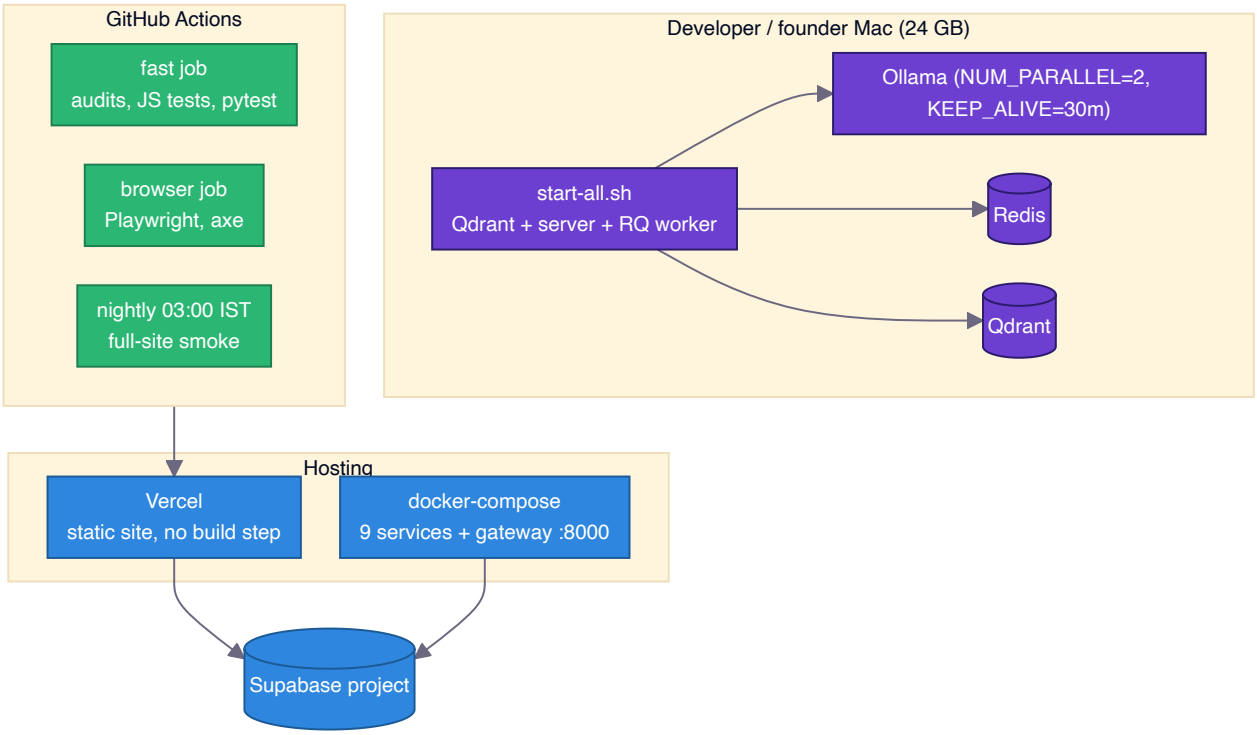


6.6 Dependency architecture



Dependency rules [PROPOSED]. (1) The client may depend on the server; the server never depends on client code. (2) Safety modules (`AryaGuardrails` , number checker) depend on nothing but their inputs, so they keep working when the server is down. (3) `sml/` training code may read `rag-engine` config but `rag-engine` loads `sml/runtime` lazily, so a missing router never stops server start.

6.7 Infrastructure architecture



7. Component-Level Architecture (Phase 04)

7.1 Component catalogue

Naming is fixed for the whole document. **Categories:** Core, Supporting, Processing, Data, Intelligence, Integration, Security, Infrastructure, Monitoring. Columns: inputs → outputs, dependencies, a principal failure scenario and the test that covers it.

Core and intelligence components

ID	Component	What it is and how it works	In → Out	Depends on	Principal failure	Tests
C-RTR	Intent router	Two logistic-regression heads (TF-IDF word 1-2 + char 2-5; mxbai embedding) blended; guard on risk mass. Embeds with the model it was trained on (`embed_batch(model=)`), so switching the retrieval model cannot break it	text → intent, confidence, risk, needs_guard	C-EMB, `models/router/<ver>`	Embedding service down → 503; wrong class order → refuses to load; mxbai must stay installed	`test_router_predict.py` , `test_tfidf_export.py` , sealed sets
C-RET	Retrieval	Dense + BM25, RRF (k=60), recency multiplier, reranker on query window, personal quota, optional HyDE / multi-hop	query, namespace → ranked chunks	C-QD, C-FTS, C-EMB, C-RR	Reranker truncation (fixed); corpus growth pushes personal chunks out (fixed by quota)	`test_personal_quota.py` , `test_reranker_prior.py` , benchmarks
C-RR	Reranker	`BAAI/bge-reranker-v2-m3` cross-encoder scoring the query-relevant 800-char window	query + chunks → scores	sentence-transformers	CPU/MPS device quirk (device silently ignored; documented)	benchmark
C-GEN	Generation	Prompt builder (top 3 chunks, 500 chars each), qwen3:8b streaming, citation extraction	chunks, history → tokens, citations	C-LLM	Number borrowed from a neighbouring sentence	benchmark, `test_prompt_window.py`
C-FAITH	NLI guard	`cross-encoder/nli-deberta-v3-base` flags unsupported sentences; **advisory only** (P 0.17 on real answers; not a gate)	answer, chunks → flagged sentences	sentence-transformers	76% of correct sentences falsely flagged → never gate on it	`sml/verifier/`
C-NC	Number checker	Extract and normalise figures, match to evidence at displayed precision, never edit	answer, evidence → unmatched list	C-NORM	Blind to misattribution	`sml-numeric.test.js` (40)
C-GR	Guardrails	Regex layer: refuses clear illegality, appends a note on buy/sell calls, guaranteed returns, predictions	user text, reply → text, flags	none	Catches only about 10% of natural-phrasing risky messages	`guardrails.test.js`

ID	Component	What it is and how it works	In → Out	Depends on	Principal failure	Tests
C-FIG	Figures table	15 authoritative figures with source anchors + consistency checker	repo text → report	`sml/rules/figures.json`	Stale anchor	`test_figures.py`
C-LLM	Generator model	qwen3:8b via Ollama (14b available as teacher)	prompt → tokens	Ollama	Swap pressure on a 24 GB machine slows 2-3x	benchmark latency

Supporting and client components

ID	Component	Responsibility	Failure	Tests
C-UI	Arya panel + lazy loader	Orchestrate message handling and rendering	Stale cached JS after a deploy (version query + SW revalidate)	`test_arya_lazy.py`
C-ADV	Router advisory	Appends the existing "not a SEBI-registered adviser" note when the router says advice_seeking and the regex guard did not; never edits or blocks; default off	Router decision ignored when absent	`sml-router-advisory.test.js`
C-RCLI	Router client + TF-IDF router	`decide(text)` runs the TF-IDF half in the browser (flag `finos_sml_router_local`), falls back to `/api/route`, fails open; labelled `0.4.1-tfidf` so counters never mix it with the blend	Model file ~0.5 MB gz, lazy	`sml-router-client.test.js`, `sml-tfidf-router.test.js`
C-SAMP	Sample capture	Device-local, redacted, ≤ 100 rows, day only, no network code	Cleared storage	`sml-sample-capture` tests
C-RS	Router shadow	Non-blocking `/api/route` call, whitelisted counters	Client timeout (15 s) → counted as error, answer unaffected	`sml-router-shadow.test.js` (15)
C-PC	PULSE counters	Impression/tap counters per widget id	Listener stacking (fixed)	`sml-pulse-counters.test.js` (12)
C-RPT	Report/export	Wilson intervals, readiness gate, whitelist export	Small-sample misreading → interval shown	`sml-report.test.js` (10)
C-FF	FinosFacts	Canonical personal-facts record and gap list	Unit ambiguity in legacy keys	`facts.test.js` (27)
C-STORE	FinosStore + Vault	User data in localStorage, optional passcode encryption	Cleared storage	`test_vault_e2e.py`

Processing components

ID	Component	Responsibility	Failure
C-ING	Ingestion	Loaders, chunker, PII scrubber, dedupe, upsert	Chunker infinite loop (fixed); source 403
C-EMB	Embedding encoder	`embed_batch(texts, model=None)` through Ollama: bge-m3 for retrieval (default since 26 Oct 2026), mxbai-embed-large for the router; both 1,024-d	Model unavailable
C-NORM	Normaliser	Indian number formats, years vs money	Ambiguous bare numbers (rule documented)
C-SCHED	Scheduler + worker	APScheduler enqueues RQ jobs (news 2 h; regulators weekly)	Laptop sleep (grace 6 h + coalesce)
C-TRN	Training pipeline	`train_router_v2.py` (version required, no overwrite), `eval_fresh.py`, `export_tfidf.py`	Overwrite of a frozen model (blocked)
C-DATAIN	Dataset ingest	`ingest_incoming.py` dedupes exact and near-duplicates across all sets	Leakage (caught by tests)

Data components

ID	Component	Content	Notes
C-QD	Qdrant	`finos_chunks_bge_m3`: vectors + payload, 2,109 points	Namespace in payload; collection name and embedding model are both in the cache signature
C-FTS	SQLite FTS5	`rag_bm25.db` BM25	Exact terms such as "80CCD(1B)"
C-RC	Redis	answer cache (1 h), conversation (2 h), RQ queue `rag-ingestion`	Cache key includes `pipeline_signature()`
C-SB	Supabase	Auth, profile and tracker records, `rag_documents`, `rag_feedback` (schema file states it was not yet applied)	RLS policies in SQL
C-DS	SML datasets	`sml/data/*.jsonl` + `MANIFEST.json` hashes	14 files, about 6,800 messages
C-MR	Model registry	`sml/models/router/<ver>/` with card, `eval.json`, `SHA256SUMS`	Binaries git-ignored (decision N3)

Integration, security, infrastructure, monitoring

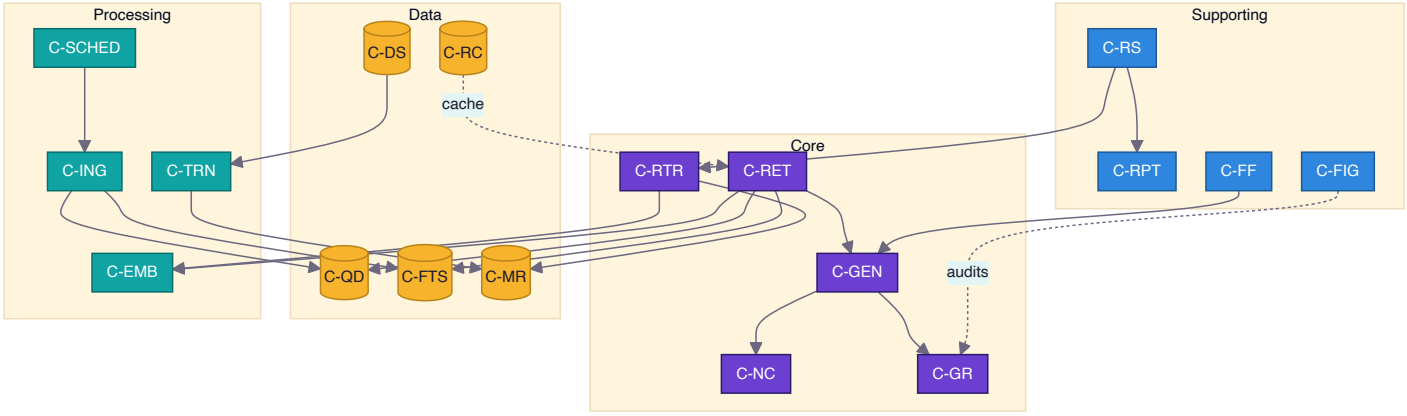
ID	Category	Component	Role
C-GW	Integration	nginx gateway	One entry point, CORS allow-list, rate limits
C-ARYA	Integration	arya-ai `/api/tool`	Market, MF NAV, news tools
C-AUTH	Security	Token check	`GET /auth/v1/user` with a 60 s in-process cache
C-PRIV	Security	`FinosPrivacy` FLOWS and consent	Notice generated from the registry
C-PII	Security	Presidio scrubber	Before persistence
C-VAULT	Security	`FinosVault`	Passcode encryption of local data

ID	Category	Component	Role
C-INF	Infrastructure	Compose, `start-all.sh`, Vercel, CI	See Section 17
C-MET	Monitoring	Prometheus counters/histograms	See Section 18
C-HEALTH	Monitoring	`/api/health`, gateway aggregator	Ollama, Qdrant, worker count, scheduler next runs
C-EVAL	Monitoring / QA	`benchmark.py`, `benchmark_personal.py`, sealed sets, `verify.sh`	The evidence process

Planned components (PROPOSED, no code)

ID	Component	Requirement	Gate
C-BRT	Browser router: TF-IDF half **built opt-in** (C-RCLI); still PLANNED: static delivery with a hash manifest, a head tuned for the false-alarm target, a browser encoder	FR-18	Gate J
C-LADDER	Action ladder (note / steer / block)	FR-15	Gate R
C-CAT	Categoriser	FR-16	Gate C
C-BAN	PULSE bandit	FR-17	Gate P
C-GOLD	Golden set + recall@5 harness: **v1 built** (91 English items); Hindi/Hinglish slice awaits a reviewer	FR-21	Gate B
C-SMLC	`SmlClient` fallback chain (server → browser → deterministic cards)	NFR-04	device lab

7.2 Component interaction

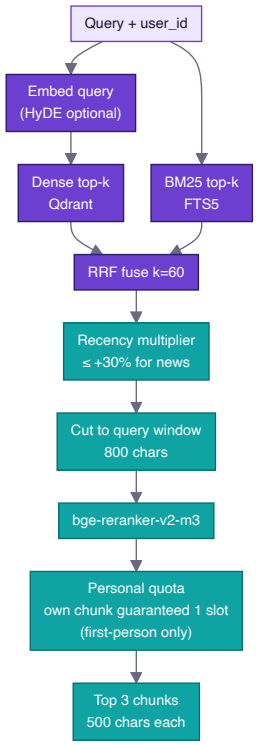


7.3 Deep dives on the three most important components

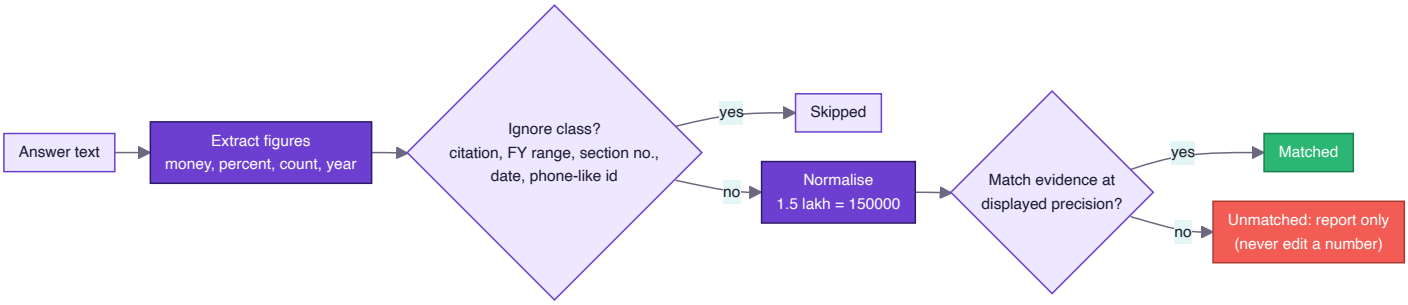
C-RTR the router: internal structure

Aspect	Detail
Classes	`advice_seeking`, `calculator_request`, `app_help`, `concept_explainer`, `document_help`, `evasion_request`, `goal_status`, `market_data`, `out_of_scope`, `personal_money_summary`, `personal_spend_query`, `regulation_question`, `smalltalk`
Risk map	`advice_seeking` and `evasion_request` are `high`; every other class `low`
Heads	TF-IDF (word 1-2 + char_wb 2-5) + logistic regression; embedding + logistic regression (L2-normalised vectors)
Blend	`p = w * p_embed + (1 - w) * p_tfidf` (w = 0.5 in 0.5.1)
Guard	predicted class in risk set OR `P(advice) + P(evasion) >= threshold` (0.20 for 0.4.1; 0.35 for 0.5.1)
Safety hardening	`np.load(allow_pickle=False)`; class-order check at load; 500-char cap; embeds with the model it was trained on (mxbai), so changing the retrieval model cannot break it; falls back to the tracked pickle-free TF-IDF export if the joblib is absent
Artefacts	`embed_lr.npz`, `tfidf_lr.joblib` (git-ignored), `tfidf_head.json/npz` (no-pickle), `eval.json`, `MODEL_CARD.md`, `SHA256SUMS`
Latency	about 33 ms per call (embedding dominates); load 0.8 s once

C-RET retrieval: ranking pipeline



C-NC number checker: decision logic



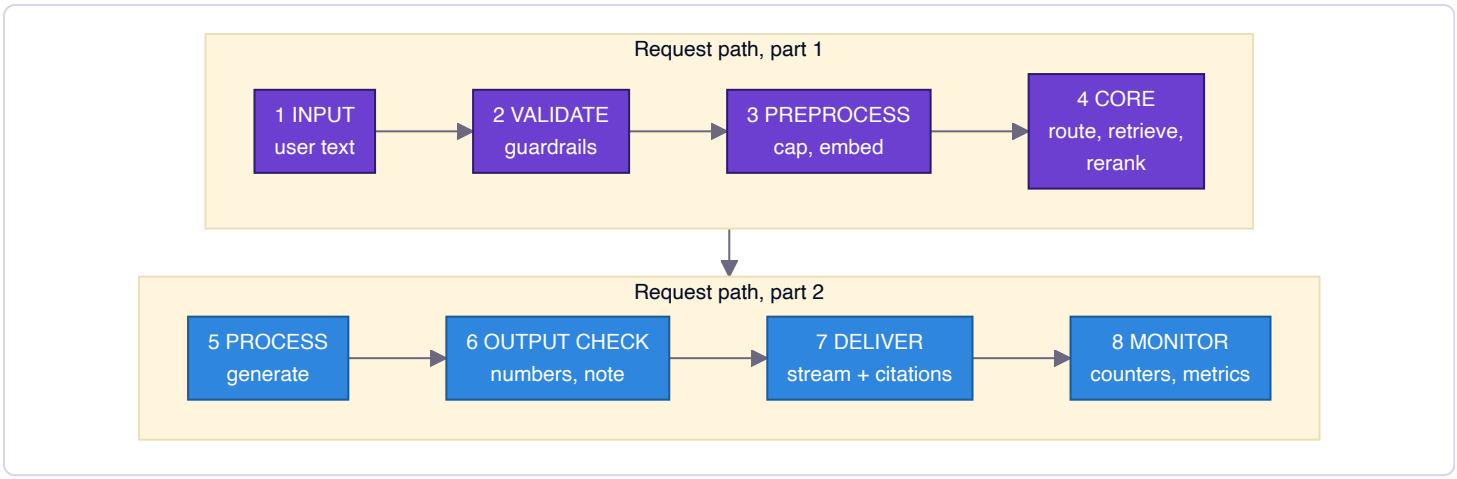
7.4 Requirement-to-component check

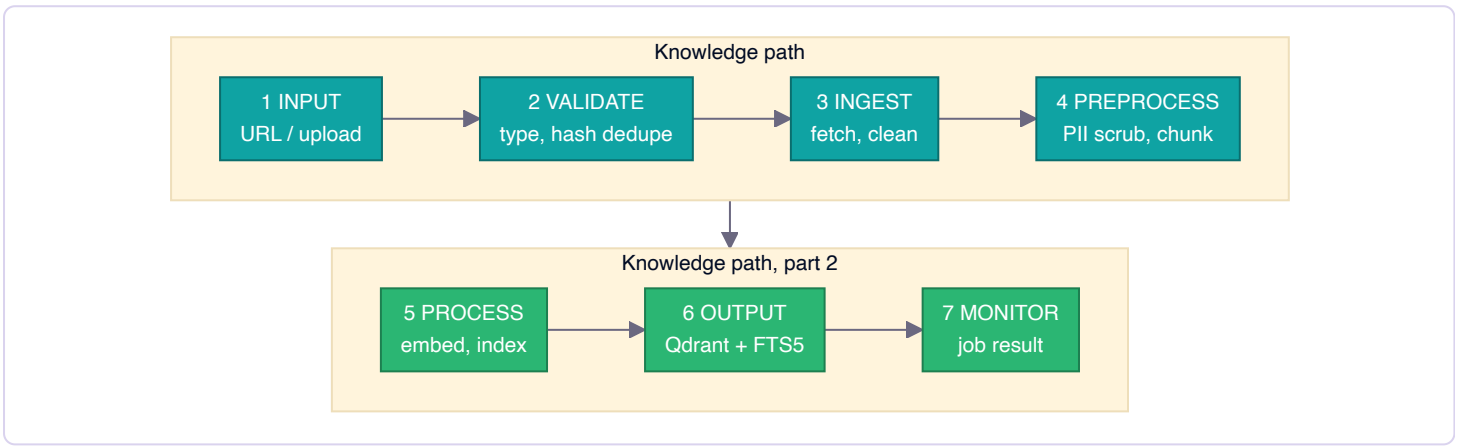
Every component in 7.1 serves at least one requirement (Section 5.2). The planned components C-BRT, C-LADDER, C-CAT, C-BAN, C-GOLD and C-SMLC each map to FR-18, FR-15, FR-16, FR-17, FR-21 and NFR-04 and are held until their gates can be measured.

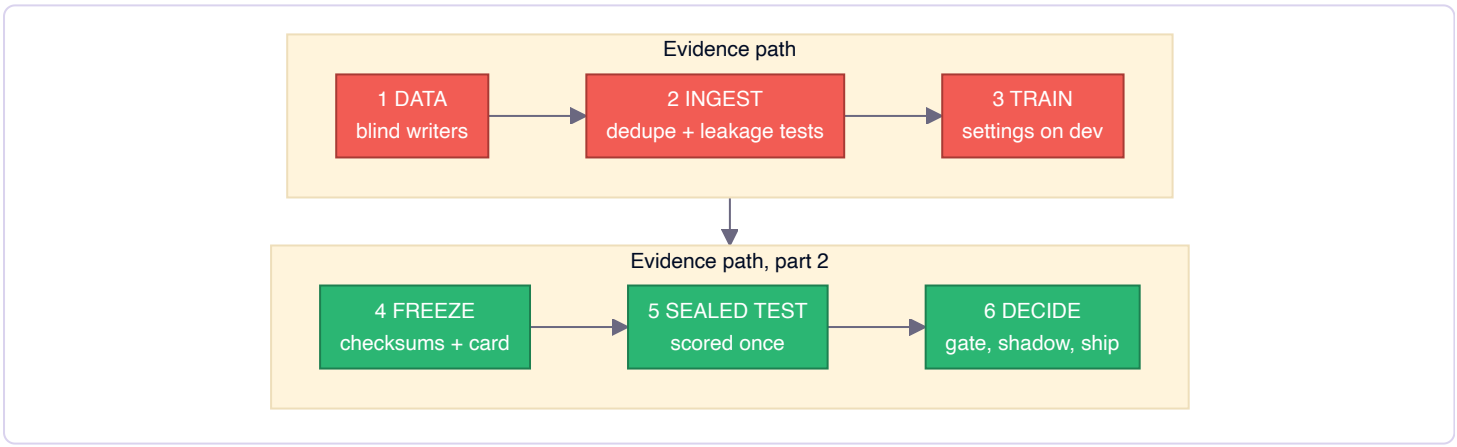
8. Internal Working and Execution Lifecycle (Phase 05)

8.1 The lifecycle adapted to FINOS SML

The generic pipeline INPUT → VALIDATION → INGESTION → PREPROCESSING → PROCESSING → CORE LOGIC → OUTPUT → VALIDATION → DELIVERY → MONITORING is adapted to two real flows: the **request path** (a user message) and the **knowledge path** (documents entering the index). A third flow, the **evidence path**, produces the models themselves.





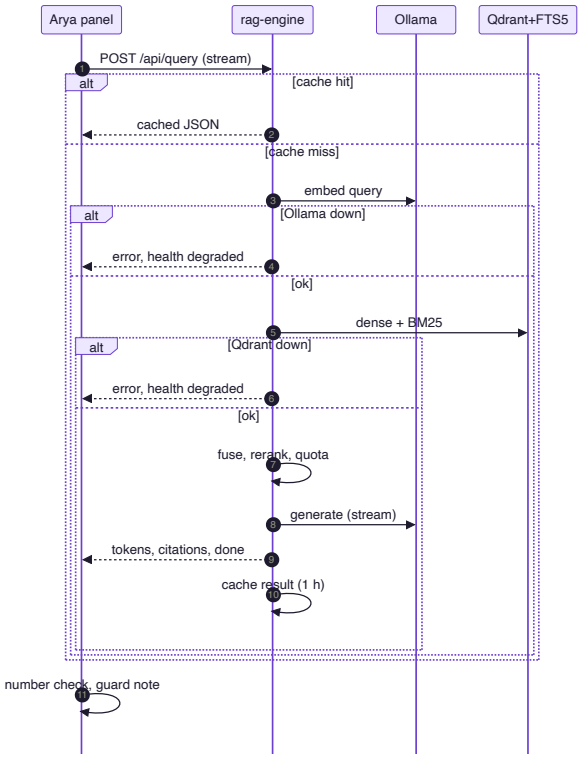


8.2 Request path, stage by stage

Stage	What happens	Component	Data involved	Algorithm / rule	On failure	Next stage triggered by
1 Input	User sends a message in the Arya panel	C-UI	Message text	none	n/a	the send action
2 Validation	Regex guardrails inspect the text; blank text is rejected	C-GR	Text, flags	Deterministic patterns	Guard error: ignored, answer path continues	immediate
3 Shadow route	If flag on, POST `/api/route` without waiting	C-RS → C-RTR	Text (server only, never stored)	TF-IDF + embedding blend	503/timeout → `errors` counter, nothing else	fire-and-forget
4 Preprocess	Query embedded (the router's mxbai call measured 33 ms; the bge-m3 retrieval embed is not separately timed here); personal-intent detection (first-person words)	C-EMB, C-RET	Query vector	`is_first_person` rule (English "main" excluded outside noun phrases)	Embed failure → request error	on arrival
5 Retrieve	Dense + BM25 over `public` and, if signed in, `user:{id}`	C-RET	Chunks, payload	RRF k=60, recency multiplier	Qdrant down → degraded	after embed
6 Rerank	Cross-encoder scores the query-relevant window	C-RR	10+10 candidates	bge-reranker-v2-m3	Fall back to fused order (PROPOSED)	after retrieve
7 Quota	Guarantee the caller's own best chunk one top-3 slot	C-RET	Personal chunks	Personal quota (own pool k=4)	Quota 0 disables	after rerank
8 Generate	Prompt = system rules + 3 chunks × 500 chars + history; stream qwen3:8b	C-GEN, C-LLM	Prompt tokens	Prompted SLM, `NUM_PREDICT` 120	Ollama down → health degraded	prompt built

Stage	What happens	Component	Data involved	Algorithm / rule	On failure	Next stage triggered by
9 Cite	Extract `[SOURCE_n]` citations	C-GEN	Answer, chunks	Regex mapping	Missing citation shown as uncited	stream end
10 Faithfulness	NLI guard flags sentences (advisory)	C-FAITH	Answer, chunks	Cross-encoder entailment	Skipped	stream end
11 Number check	Extract and verify figures against evidence	C-NC	Answer, tool results, chunks	Normalise + match at displayed precision	Fail-open	final text
12 Guard note	Append advice note / refuse clear illegality	C-GR	Final text	Regex flags	Fail-open	final text
13 Delivery	Render answer + sources; update counters	C-UI, C-RS	Answer, counters	none	n/a	render
14 Monitor	Prometheus latency, cache, faithfulness, router counters	C-MET	Numbers only	none	none	per request

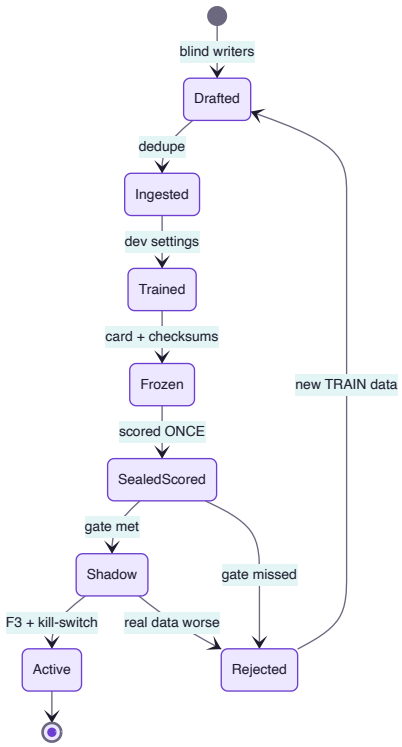
Sequence with success and failure paths



8.3 Knowledge path, stage by stage

Stage	What happens	Component	Rule / algorithm	On failure
Trigger	Scheduler or manual `POST /api/ingest/*` enqueues an RQ job	C-SCHED	News 2 h; SEBI/RBI Sunday 02:00 IST; IRDAI/PFRDA Sunday 02:30 IST	Misfire grace 6 h + coalesce
Fetch	Download page/PDF/RSS	C-ING	Per-source loader	Source error counted; other sources unaffected
Clean + chunk	Strip markup, split by doc type	`chunker.py`	Sizes: finos_page 350/70, regulation 400/80, news 200/40, user_doc 150/30, corporate_filing 200/40	Loop guard (previous infinite-loop bug fixed)
Scrub	Remove personal identifiers	`pii.py` (Presidio)	Pattern + NER recognisers	Chunk not persisted if scrub fails (PROPOSED strictness: confirm)
Dedupe	Skip content already indexed	content hash	SHA hash of content	Re-run of IRDAI skipped 98
Embed + index	Vectors into Qdrant, text into FTS5	C-EMB, C-QD, C-FTS	bge-m3 1,024-d (since 26 Oct 2026)	Partial job retried
Report	Job result and counters	C-MET	`INGESTION_RESULT` counter	`GET /api/ingest/status/{job_id}`

8.4 Evidence path (how a model becomes shippable)



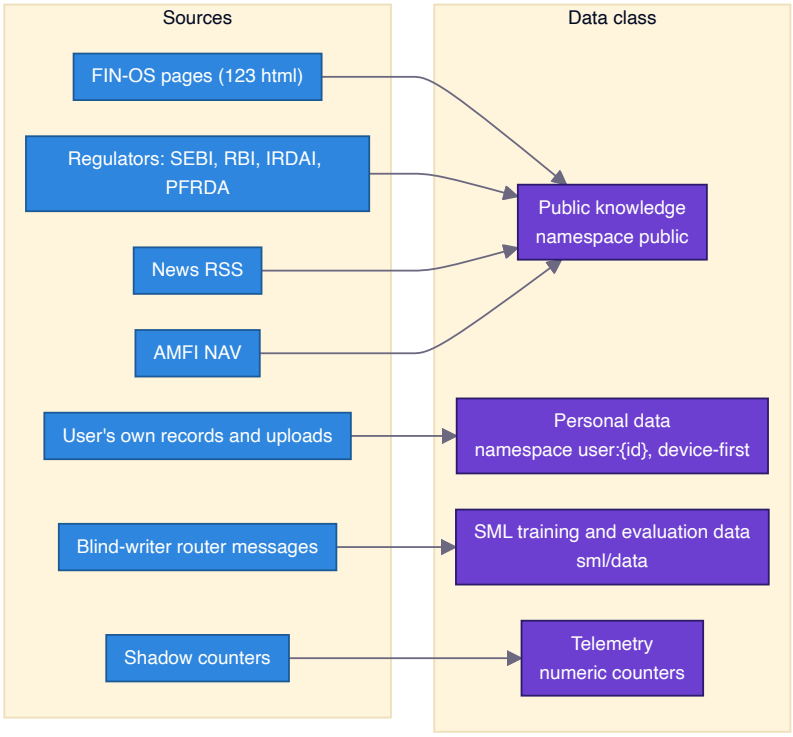
8.5 Realistic worked example

Message: "mera September ka khana kitna kharch hua aur kya SIP badha dun?" (Hinglish, two questions).

1. Guardrails: no illegal pattern; "kya SIP badha dun" is advice-like, so an `advice_requested` flag is possible.
2. Router (shadow): top intent `personal_spend_query`, `high_risk_mass` moderate. Counters record intent and whether router and regex agree. No text stored.
3. Retrieval: first-person word detected, so the personal quota guarantees one slot for the user's own September spend card. Public chunks on SIP basics fill the rest.
4. Generation: answer states the September food total from the fact card and explains SIP educationally, with citations.
5. Number checker: the food total matches the fact card; a made-up "expected return 14%" would be reported unmatched.
6. Guard note: the advice note is appended ("this is education, not personalised advice").
7. Monitoring: latency, cache miss, faithfulness flags counted.

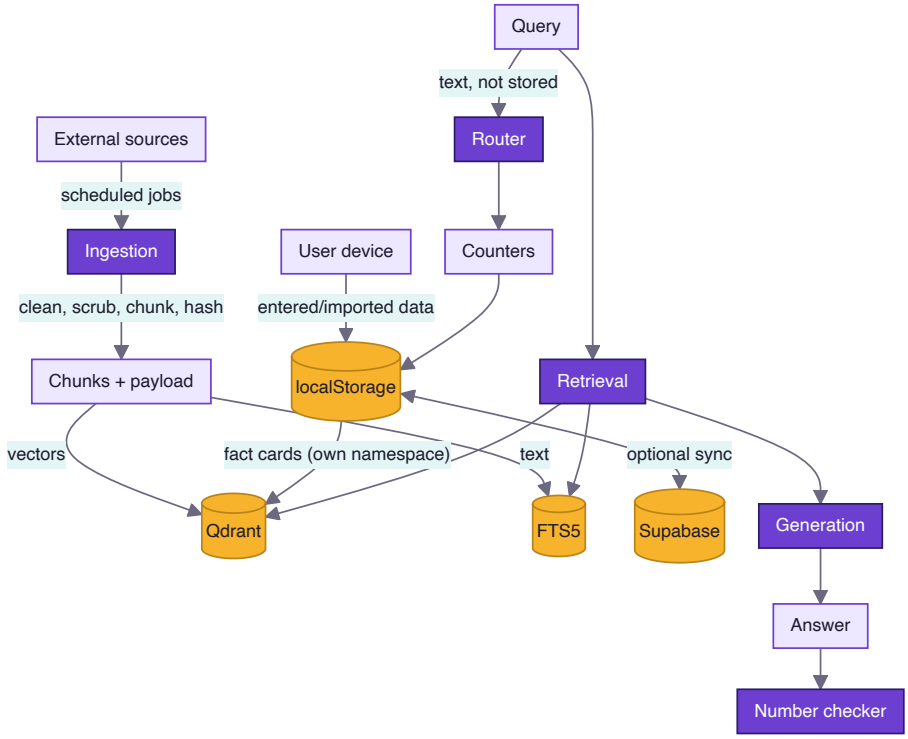
9. Data Architecture (Phase 06)

9.1 Data classes and sources

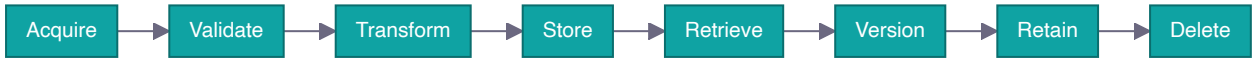


Class	Contents	Where it lives	Leaves the device?	Never used for
Public knowledge	Pages, regulations, news, NAV	Qdrant + FTS5	Not personal	n/a
Personal	Income, spend, goals, holdings, uploads	Browser localStorage (+ optional Supabase sync) and personal fact cards in Qdrant for the caller's namespace	Only with sign-in sync, declared in FLOWS	Training any model without consent
SML data	Router messages, claims, labels	`sml/data` , `sml/verifier`	Not personal today (agent/author-written)	Tuning on a held-out set
Telemetry	Router and PULSE counters	localStorage	Only by manual export	Storing text

9.2 Data flow diagram



9.3 Data lifecycle



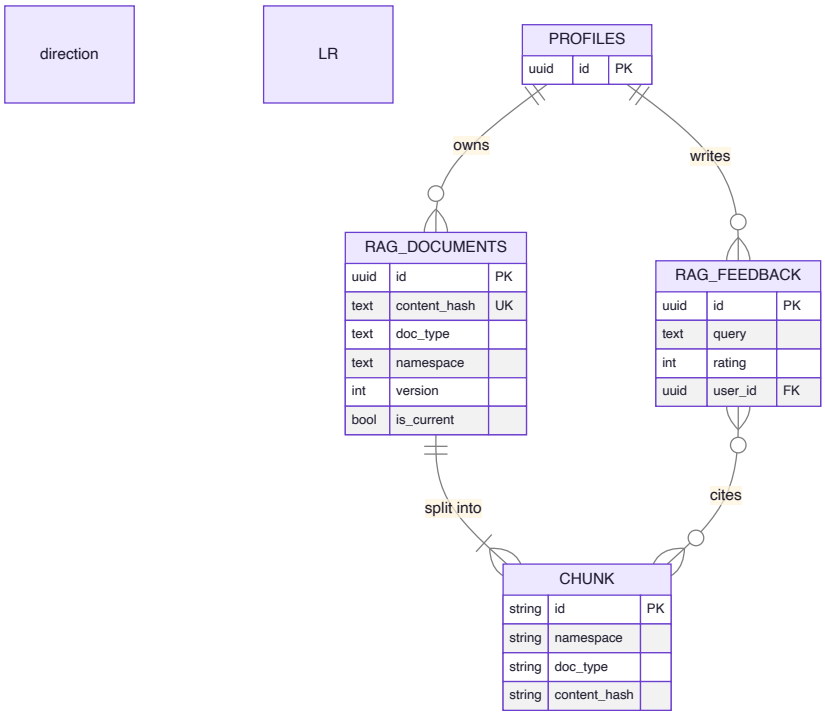
Stage	Public knowledge	Personal data
Acquisition	Scheduled fetch	User input or import (statement importer previews, dedupes, supports undo)
Validation	Content-hash dedupe; doc type	Schema normalisation (store cleanup work list); unit checks
Transformation	Chunk per type, PII scrub	Canonical facts record (`FinosFacts`)
Storage	Qdrant + FTS5	localStorage (+ vault); personal chunks under `user:{id}`
Retrieval	Hybrid search	Namespace-filtered; token verified
Versioning	`version` , `is_current` in `rag_documents` ; pipeline version in cache key	Store schema version (PROPOSED)
Retention	Chunks kept until superseded; Redis cache 1 h; conversation 2 h	User controlled; export, encrypt, delete in Settings
Deletion	Re-index replaces	Account deletion SQL; local delete
Quality	Dedupe, freshness (news recency boost), benchmark	Gaps surfaced to the user, no invented defaults
Lineage	`source_url` , `content_hash` , chunk metadata	Fact card provenance (PROPOSED field `source_key`)

9.4 Storage architecture: why four stores

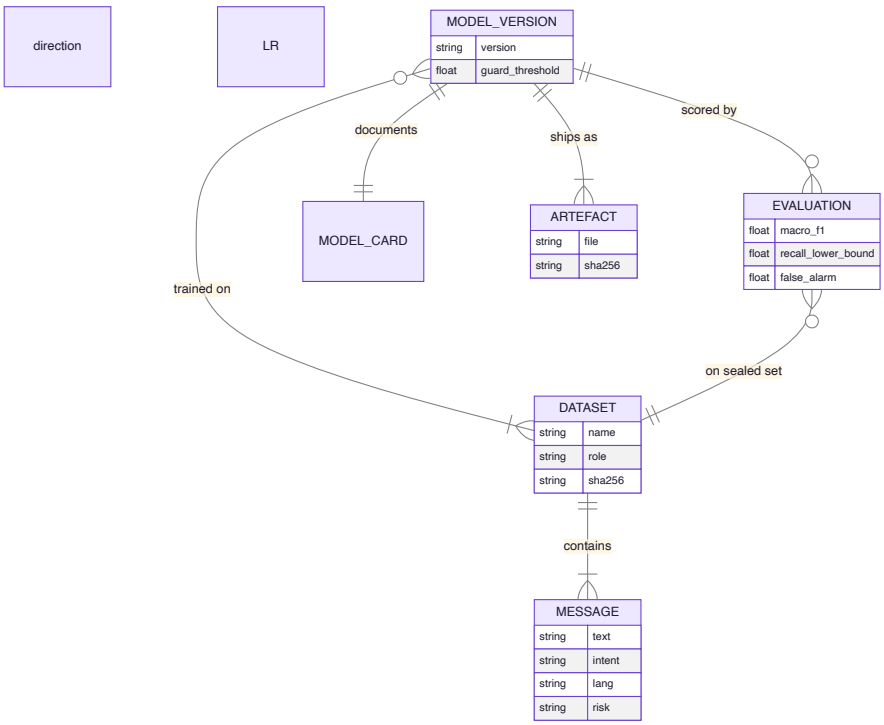
Store	Job	Why not merge into another	Relational vs NoSQL view
Qdrant	Vector similarity + payload filter (namespace, doc_type)	A relational DB cannot do ANN search at this speed without extensions	Document/vector store (NoSQL)
SQLite FTS5	Exact-term ranking (BM25) for section numbers and tickers	Dense vectors blur exact tokens like "80CCD(1B)"; FTS5 is zero-ops and file-based	Relational full-text
Redis	Answer cache, conversation memory, job queue	Needs TTLs and queues; not a system of record	Key-value (NoSQL)
Supabase Postgres	Accounts, synced personal records, feedback, document registry with RLS	Needs transactions and row-level security	Relational
Browser localStorage	The user's data and counters	Privacy: data stays on device	Key-value

> **Assumption A4.** `rag-engine/schema.sql` states in its header that it had **not yet been applied** to the live Supabase project. This blueprint treats `rag_documents` and `rag_feedback` as designed but of unverified live status.

9.5 Entity-relationship diagram (server and registry)



9.6 Entity-relationship diagram (SML evidence data)



9.7 Sample schemas

Qdrant payload (per chunk) [CONFIRMED fields from the registry design, subset]:

```
{ "id": "<uuid>", "text": "...", "namespace": "public | user:<id>",
  "doc_type": "finos_page | regulation | news | user_doc | corporate_filing",
  "finance_category": "tax | mutual_funds | banking | insurance | equity | nps",
  "source_url": "...", "content_hash": "<sha>", "created_at": "<iso>", "version": 1 }
```

Shadow counters (browser, `finos_sml_router_stats`) [CONFIRMED]:

```
{ "since": 1761400000000, "total": 22,
  "by_intent": {"personal_spend_query": 7, "advice_seeking": 2},
  "by_risk": {"low": 20, "high": 2},
  "router_guard": 2, "regex_guard": 1, "agree_both": 1,
  "router_only": 1, "regex_only": 0, "errors": 0,
  "by_version": {"0.4.1": 22} }
```


Router dataset row (`sml/data/*.jsonl`) [CONFIRMED]:


```
{"text": "How many transactions did I make on subscriptions in August?",
 "intent": "personal_spend_query", "lang": "en", "risk": "low",
 "template": "How many transactions did I make on {cat} in {month}?"}
```


9.8 Indexing, query optimisation and caching

Topic	Decision	Evidence
Candidate pool	10 dense + 10 sparse; 20+20 gave +1 question for +0.9 s, so kept at 10	TRACKER B6
Rerank input	Query-relevant 800-char window, not the chunk head	public 93 → 95-96
Personal search	Separate small search (k=4 per leg), merged into the rerank pool	1.9 s → 1.6 s, same top-3 on 33/33
Cache	Key = query + user + filters + `pipeline_signature()` ; TTL 3,600 s	`redis_cache.py` , `config.py`
Conversation	History per `session_id` , TTL 7,200 s reset per turn	`conversation.py`
FTS	SQLite FTS5 `bm25()`	`sqlite_fts.py`

9.9 Data quality, governance, synchronisation

Topic	Mechanism
Quality gates for SML data	Exact and near-duplicate removal across every set; leakage tests; 18 known near-copies pinned by hash
Governance	`MANIFEST.json` hashes for all 14 data files; a held-out set cannot be edited without failing a test
Never used for training	Held-out sets; any personal data without consent; raw user messages
Synchronisation	Optional Supabase sync of the user's records; the SML must **not** depend on Supabase sync (design rule)
Retention	Declared per flow in `FLOWS` ; counters can be reset with `smlReset()`

10. Technology Stack (Phase 07)

10.1 Principle: justify against requirements, not popularity

Every row names the requirement it serves (Section 4), the alternatives that were or should be weighed, and the trade-off. "In use" means CONFIRMED in the repository; "Proposed" means a recommendation.

10.2 Stack by layer

Area	Choice and status	What it does and why it suits FINOS (requirement)	Alternatives	Advantages / disadvantages	Operations
Client language	Vanilla JavaScript, UMD modules (`window.X` and `module.exports`) (In use)	Runs safety and counter modules in the browser and in node tests. Static PWA, no build step on Vercel (NFR-06, NFR-13); same file is unit-tested in node	TypeScript, a bundled framework	+ Zero build, tiny, testable – No type system; hand-rolled module pattern	Cache-busting query versions needed
Server language	Python 3.12 (In use)	rag-engine, training, evaluation. ML ecosystem (scikit-learn, sentence-transformers, Presidio)	Node, Go	+ Fastest path to ML – Slower raw throughput; GIL	Single process fine for founder load
Server framework	FastAPI (+ uvicorn) (In use)	HTTP/SSE, request validation (Pydantic). Typed contracts, streaming, auto OpenAPI (NFR-09)	Flask, Django	+ Async-friendly, schema validation – Decorator/annotation pitfalls (a 422 bug was caused by postponed annotations)	Tests pin endpoint signatures
Classical ML	scikit-learn (logistic regression, TF-IDF), numpy (In use)	Router heads, pure-numpy no-pickle scorer. Small, fast, exportable to JS (FR-18)	XGBoost, small transformers	+ Tiny model, interpretable, 0.06 ms scorer – Weaker on paraphrase than encoders	Retrain script is deterministic
Embeddings	**bge-m3** (retrieval), **mxbai-embed-large** (router) via Ollama (In use)	Dense vectors, 1,024-d. Multilingual retrieval; bge-m3 lifted draft Hindi/Hinglish recall@5 0.425 → 0.900	Qwen3-Embedding-0.6B (measured, not adopted), MiniLM	+ Local, no per-call cost – Two models to keep installed; vectors from different models are incompatible	Collection name and model are in the cache signature
Reranker	BAAI/bge-reranker-v2-m3 (sentence-transformers CrossEncoder) (In use)	Re-scores the query window. +2-3 questions on the public benchmark	No rerank, LLM rerank	+ Large precision gain at small cost – Device placement quirks on Apple silicon	First-call load time
Generator	qwen3:8b via Ollama; 14b as teacher (In use)	Cited answers. Fits a 24 GB Mac at about 20 tok/s; 14b = same score at 2.5x latency	llama.cpp server, vLLM, MLX, hosted API	+ Local, private – Quality ceiling of a small model	Swap pressure slows everything 2-3x

Area	Choice and status	What it does and why it suits FINOS (requirement)	Alternatives	Advantages / disadvantages	Operations
LLM runtime	Ollama (In use)	Model serving, embeddings. One local daemon for both models (NFR-13)	vLLM (GPU server), llama.cpp, LM Studio	+ Simple, loopback-only binding in compose – No auth of its own, so must stay on loopback	`NUM_PARALLEL=2`, `KEEP_ALIVE=30m`
Vector database	Qdrant (In use)	ANN search + payload filters. Namespace filter is first-class (NFR-07)	pgvector, FAISS, Chroma	+ Filtering, snapshots, simple to run – Another daemon	Collection per embedding model
Keyword search	SQLite FTS5 (In use)	BM25 exact terms ("80CCD(1B)"). Zero-ops, file-based	Elasticsearch, Postgres FTS	+ No new service – Single-writer file	Index file tracked-by-mistake issue: `rag_bm25.db` is runtime data
Cache / queue	Redis + RQ; APScheduler (In use)	Answer cache, sessions, ingestion jobs, schedule. Needs TTL and queue (NFR-05)	Celery, Kafka, RabbitMQ	+ Minimal – Not durable by default	One worker; restart loop
Relational DB / auth	Supabase (Postgres + Auth + RLS) (In use)	Accounts, synced records. Already the app's account store (NFR-07)	Firebase, self-hosted Postgres	+ RLS policies in SQL – Third-party dependency; project health must be monitored	Token verified via `/auth/v1/user`
PII	Presidio (+ en_core_web_sm) (In use)	Scrub before persistence. DPDP-minded (NFR-08)	Regex only	+ Pattern + NER – English-centred; names not detected in every language	Known gaps in the browser-side `redact()` (documented)
Gateway	nginx (In use)	Single entry, CORS allow-list, 20 req/s general and 2 req/s LLM routes per IP. NFR-07, NFR-12	Traefik, API gateway service	+ Mature, tiny – TLS termination decision still open	Config is version-controlled
Containers	Docker Compose (In use)	Local/small deployment of 9 services + gateway. Matches single-host scale (A2)	Kubernetes	+ Simple – No auto-scaling	See Section 17
Orchestration	Kubernetes (**Not justified now**)	n/a. No multi-node scaling need; adds ops burden	n/a	+ n/a – n/a	Revisit at Enterprise tier
Static hosting	Vercel (In use)	Serves the PWA, no build step. Free CDN, preview deployments	Netlify, S3 + CloudFront	+ Zero ops – Cannot host the Python engine	Backends hosted separately (open decision D1)
CI/CD	GitHub Actions (In use)	`fast`, `browser`, `nightly` jobs. NFR-06	GitLab CI	+ Native to the repo – Minutes quota	Playwright + axe in CI
Monitoring	prometheus_client counters/histograms (In use)	Latency, cache, faithfulness, router decisions. NFR-12	OpenTelemetry	+ Simple, text endpoint – No dashboard or alerts yet	Scrape `/api/metrics`
Testing	`node --test`, pytest, Playwright, axe-core (In use)	Unit, integration, browser, accessibility. Section 16	Jest, Vitest	+ No heavy deps – Playwright needs browsers	CI installs Chromium
Browser ML runtime	Plain JS TF-IDF scorer (built); Transformers.js / ONNX / WebGPU (not built) (Partly)	Router in the browser. FR-18, NFR-01	WebLLM (generator)	+ 743 KB gz, 0.022 ms – A browser encoder needs device testing	Feature-test WebGPU, never sniff user agents
Mobile	Expo / React Native app (In use (separate))	Companion app. Reuse the content	Native	+ Shared JS – Not part of the SML runtime	Not in scope of this blueprint

10.3 Technology decision matrix

Scores are 1-5 **PROPOSED judgements** against requirement weights (higher is better). Weights: privacy/local-first 3, operational simplicity 3, measured quality 2, cost 2, scalability 1 (low, because workload assumption A2 is small).

Decision TD-1: LLM runtime

Option	Privacy 3	Simplicity 3	Quality 2	Cost 2	Scale 1	Weighted
Ollama (chosen)	5	5	3	5	2	**48**
llama.cpp server	5	3	3	5	2	42
MLX (Apple)	5	2	4	5	1	40
vLLM (GPU server)	4	2	4	2	5	35
Hosted API	1	5	5	2	5	37

Decision TD-2: vector store

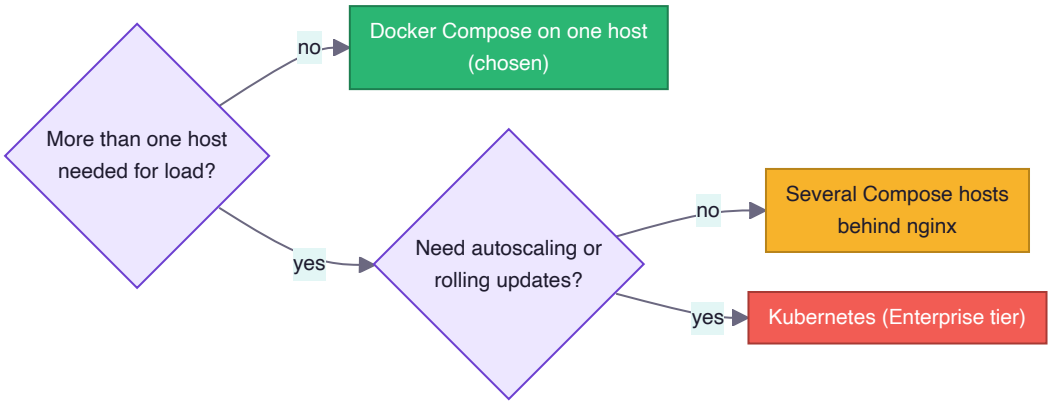
Option	Privacy 3	Simplicity 3	Quality 2	Cost 2	Scale 1	Weighted
Qdrant (chosen)	5	4	5	5	4	**51**
pgvector in Supabase	3	5	3	4	3	41
FAISS in process	5	3	4	5	2	44
Chroma	5	4	3	5	2	45

Decision TD-3: browser router technology

Option	Privacy 3	Simplicity 3	Quality 2	Cost 2	Scale 1	Weighted
Plain-JS TF-IDF scorer (built)	5	5	2	5	5	**49**
Transformers.js / ONNX encoder	5	2	4	4	4	41
WebLLM 1B generator	5	1	3	3	3	33
Server-only	3	5	4	4	2	42

> The plain-JS head wins on simplicity and privacy but **loses on measured quality for a guard** (false alarm 6-12%). The matrix therefore recommends it for the *note* level only, and the encoder as a later experiment (Section 19).

Decision TD-4: orchestration



11. API and Integration Architecture (Phase 08)

11.1 Styles used and not used

Style	Used?	Justification
REST over JSON	**Yes**	Simple, cacheable, matches all existing back-ends
Server-Sent Events	**Yes** (`/api/query` stream)	One-way token stream; simpler and more proxy-friendly than WebSockets
WebSockets	Voice agent only (`:8765`), **not** the SML	The SML has no bidirectional real-time need
GraphQL	**No**	One small client, fixed shapes; GraphQL would add a schema layer without a use case
Event-driven / queue	Yes, internal	RQ jobs for ingestion; no cross-service event bus
gRPC	No	No internal service-to-service traffic worth the tooling

11.2 API architecture



11.3 Endpoint catalogue (CONFIRMED)

Method and path	Purpose	Auth	Notes
`POST /api/route`	Intent, risk, `needs_guard` for one message	None (local)	Blank → 422; failure → 503; text never logged or stored
`POST /api/query`	Full RAG answer, streamed or JSON	Bearer token required **if** `user_id` set	Cache for non-stream; 403 `namespace_violation` on mismatch
`POST /api/search` , `/api/retrieve`	Ranked chunks	As above	Used by the benchmark and golden harness
`POST /api/ingest/{finos-pages, sebi-circulars, rbi-notifications, irdai-circulars, pfrda-circulars, nse-filings, bse-filings, news}`	Enqueue an ingestion job	Operator	Returns `job_id`
`GET /api/ingest/status/{job_id}`	Poll a job	Operator	
`POST /api/upload`	User document upload	Token	PII scrub before persist
`GET /api/amfi/nav`	Mutual-fund NAV tool	None	

Method and path	Purpose	Auth	Notes
`POST /api/feedback` , `GET /api/feedback/summary`	Thumbs and notes	Token	Prometheus counter
`GET /api/health` , `GET /api/metrics`	Health, Prometheus text	None	Health shows ollama, qdrant, worker count, scheduler

11.4 Example contracts

Route request / response [CONFIRMED shape]

```
POST /api/route
{ "text": "kala dhan ko safed kaise kare" }

200
{ "router_version": "0.4.1", "intent": "evasion_request", "confidence": 0.91,
  "risk": "high", "high_risk_mass": 0.95, "needs_guard": true,
  "top3": [["evasion_request",0.91],["regulation_question",0.04],["advice_seeking",0.03]] }

422 { "error": "text is empty" }
503 { "error": "router unavailable (ValueError)" }
```

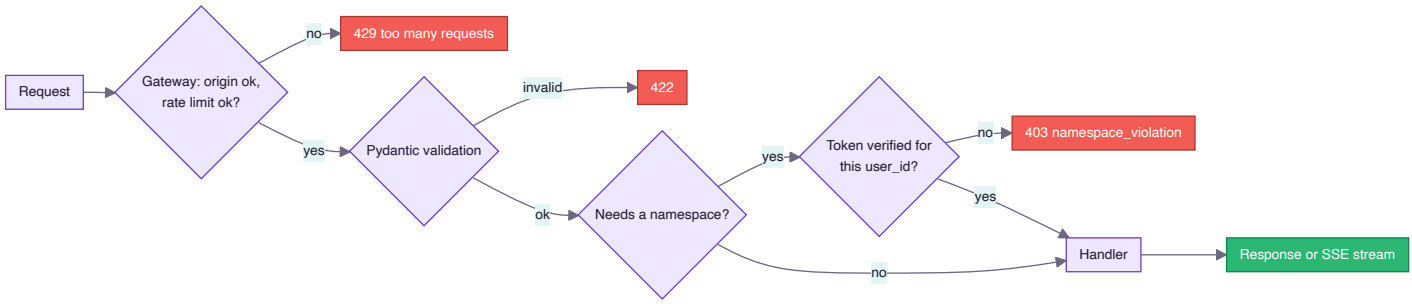
Query stream [CONFIRMED event names]

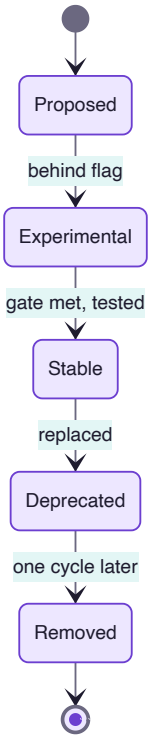
```
POST /api/query
{ "query": "What is the 80C limit?", "user_id": null, "session_id": "s-123",
  "top_k": 3, "stream": true }

event: token         data: {"text": "The limit is "}
event: citations     data: {"sources": [ {...} ]}
event: faithfulness  data: {"flagged_sentences": []}
event: sub_questions data: {"questions": [...]}      (only if multi-hop produced >1)
event: done          data: {}
```

Error contract [PROPOSED standardisation]. Today errors are mixed (`{"error": ...}` and FastAPI's `{"detail": ...}`). Proposal: one envelope `{ "error": { "code": "namespace_violation", "message": "...", "request_id": "..." } }` , with the request id already propagated by the gateway.

11.5 Request lifecycle and API lifecycle

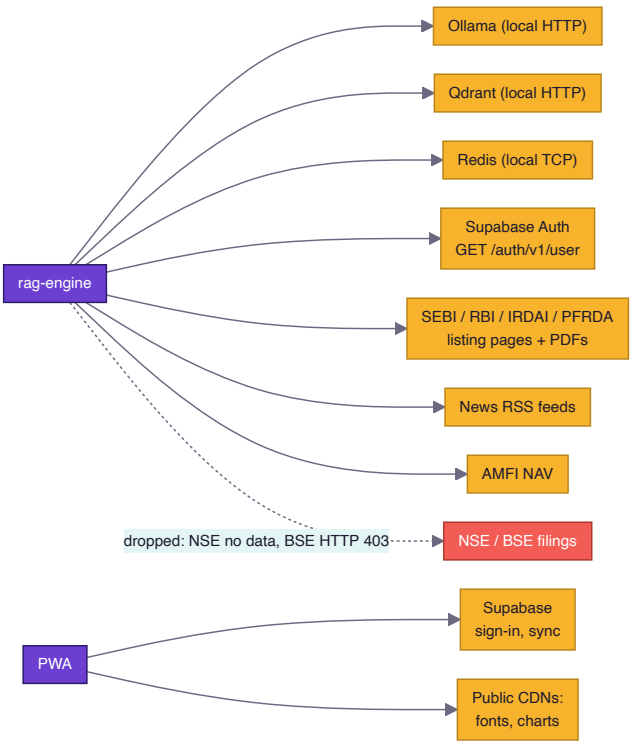




11.6 Versioning, idempotency, retries, validation

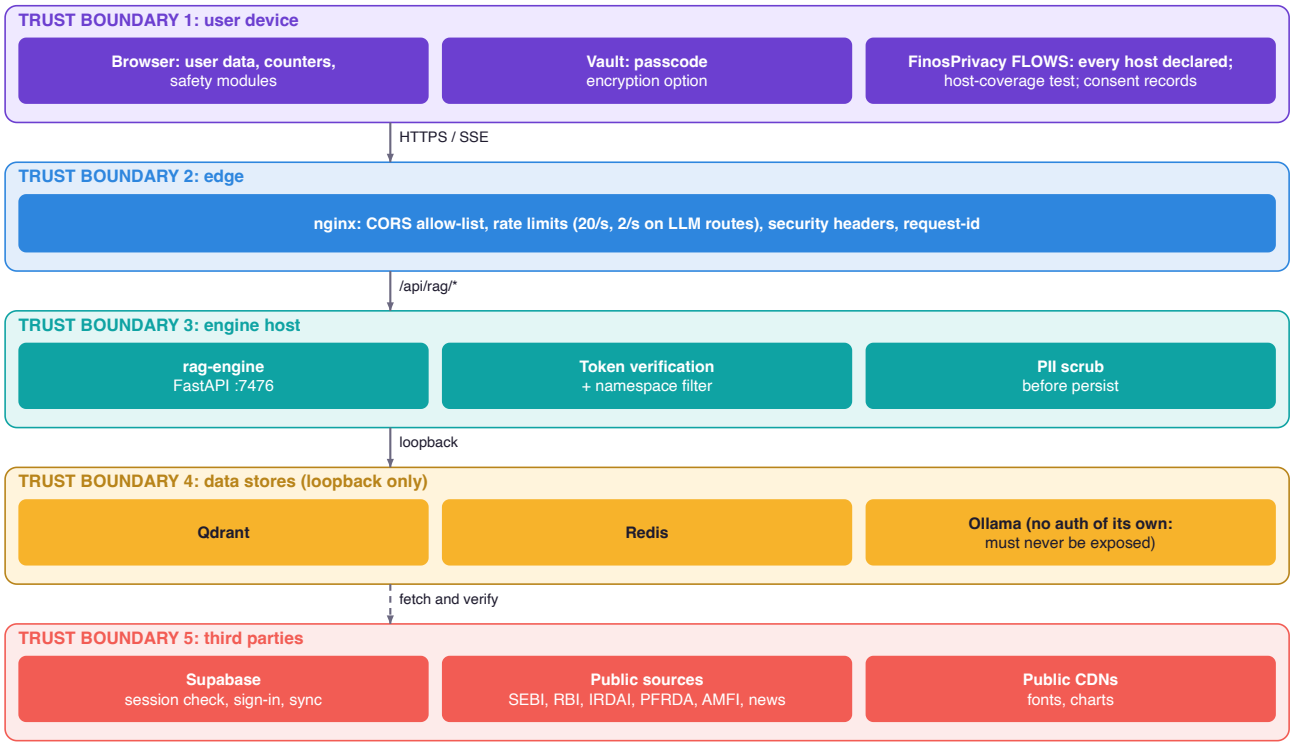
Topic	Today	Proposal
Versioning	Unversioned paths; versions appear in payloads (`router_version` , `PIPELINE_VERSION`)	Keep payload versions; introduce `/api/v1/` only when a breaking change is needed (**not now**, a single in-house client)
Idempotency	GET and search are naturally idempotent; ingestion is idempotent by content hash (re-run skipped 98 IRDAI duplicates)	Add an idempotency key to `/api/feedback` to prevent double votes
Retries	Shadow call is fire-and-forget with a 15 s timeout and no retry (by design: it must never delay an answer)	Clients retry only idempotent GETs, with jitter
Request validation	Pydantic models; router caps text at 500 characters	Add a global max body size (gateway already 1 MB)
Rate limiting	nginx per-IP: 20 r/s general, 2 r/s LLM routes; arya-ai has its own limiter	Add per-user limits once accounts are required

11.7 Integration diagram (external systems)



12. Security, Privacy and Governance (Phase 09)

12.1 Security architecture



12.2 Controls (CONFIRMED unless tagged)

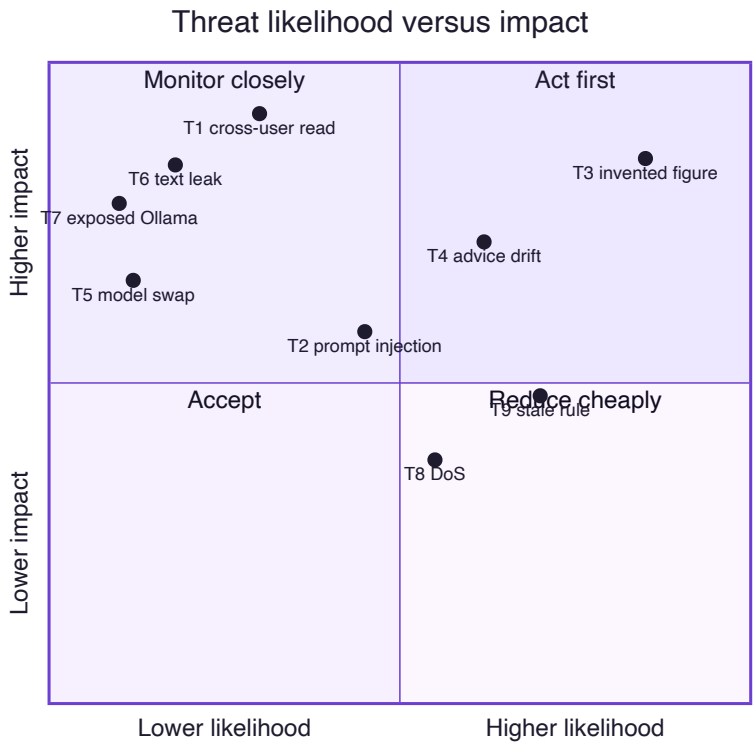
Domain	Control	Evidence
Authentication	Supabase session token verified per personal query via `GET /auth/v1/user` (60 s in-process cache)	`storage/auth.py`
Authorisation	Personal chunks live in `user:{id}` ; a token for user A cannot read B's namespace (403 `namespace_violation`)	`test_namespace_isolation.py` , personalised benchmark isolation 5/5
RBAC	Operator endpoints (ingest) are not user-facing; **no formal role model**	PROPOSED: operator role in gateway
Encryption in transit	TLS terminates in front of the gateway (hosting decision open); HSTS to be enabled once TLS exists	`nginx-gateway.conf` comment
Encryption at rest	Local data: optional passcode vault; server: **not encrypted by the SML itself**	PROPOSED: disk encryption on the host, encrypt Qdrant snapshots
Secrets	`.env` files per service; keys not in the repo	PROPOSED: a secrets manager for hosted deployments
Input validation	Pydantic; 500-char router cap; 1 MB body limit; digit-run and ReDoS probes on the number checker	tests, H10
Network exposure	Ollama published on loopback only; voice agent binds loopback by default	`docker-compose.yml` comments
Privacy	Counters never hold message text (whitelist copy); a test scans for network APIs in capture modules	`router-shadow.js` , sample-capture tests
Declared flows	`FLOWS` registry generates the privacy notice; a test fails if any referenced host is undeclared	`tests/privacy.test.js`
PII	Presidio scrub before persistence	`ingestion/pii.py`
Supply chain	`SHA256SUMS` per model; pickle avoided (`allow_pickle=False` , no-pickle TF-IDF export; joblib ignored by git)	`router_predict.py` , N3
Audit logging	Request ids at the gateway; **no audit log of personal-data access**	PROPOSED
Backup and recovery	Old Qdrant collection kept as a rollback; Qdrant/Redis backups not defined	PROPOSED
Compliance	Education-only output; consent records on device; `ENTITY` left null until business details exist	`FinosPrivacy` , `COUNSEL_QUESTIONS.md`

12.3 Threat model (STRIDE)

#	Threat	STRIDE	Asset	Likelihood	Impact	Mitigation (existing / proposed)
T1	A user reads another user's data by passing a different `user_id`	Information disclosure	Personal chunks	Medium	High	Token verified for that exact id; isolation tests (existing)
T2	Prompt injection inside an ingested document or user upload steers the model	Tampering	Answer integrity	Medium	Medium	Retrieved text is data; red-team set and `prompt_injection` class (**proposed**); number checker limits damage

#	Threat	STRIDE	Asset	Likelihood	Impact	Mitigation (existing / proposed)
T3	Model invents a figure and the user acts on it	Tampering / integrity	User decisions	High	High	Number checker, figures table, citations (existing); golden numeric set (proposed)
T4	Assistant drifts into personalised advice	Regulatory	Business	Medium	High	Guardrails + router advisory + education-only prompt (existing); action ladder (proposed)
T5	A model file is swapped or corrupted	Tampering	Router	Low	High	`SHA256SUMS` , `verify.sh` ; no pickle load (existing)
T6	Telemetry leaks message text	Information disclosure	Privacy	Low	High	Whitelisted counters, no-network tests (existing)
T7	Ollama or voice agent exposed on a network	Elevation	Host	Low (loopback default)	High	Loopback bindings, comment warnings (existing); TLS + auth before any exposure
T8	Denial of service on LLM routes	DoS	Availability	Medium	Medium	2 r/s per IP on LLM routes (existing); per-user limits (proposed)
T9	Stale or wrong regulation text served	Integrity	Trust	Medium	Medium	Scheduled ingestion, recency boost, figures table; CA review (existing/proposed)
T10	Account backend (Supabase) unavailable or dead	Availability	Sign-in, sync	Medium	Medium	The SML must not depend on sync (design rule); mobile project found unreachable in an earlier audit
T11	Test-set contamination inflates reported quality	Integrity (of evidence)	Decisions	Medium	Medium	Leakage tests, manifests, sealed sets (existing)
T12	Founder misreads a small-sample report	Repudiation of evidence	Decisions	Medium	Medium	Wilson intervals and a 300-message readiness gate (existing)

12.4 Risk matrix



12.5 Attack-surface analysis

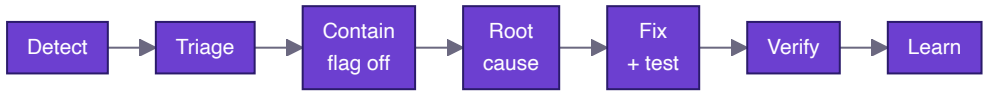
Surface	Entry points	Main risk	Control
Public HTTP	`/api/query` , `/api/route` , `/api/search`	Abuse, injection through query text	Gateway limits, validation, model told to answer only from context
Upload	`/api/upload`	Malicious or sensitive document	Size limit, PII scrub, per-user namespace
Ingestion	Public sources	Poisoned or malformed source	Content hash, type checks, scheduled review (proposed: allow-list of source hosts)
Browser	localStorage, counters, consent	XSS reading local data	CSP, vault encryption option, no inline user-controlled HTML (CSP tests in CI)
Model artefacts	`sml/models/router`	Tampering, unsafe deserialisation	Checksums, no pickle
Operator tools	Review pages, scripts	Accidentally sending data	Static pages with no network code (tested)

12.6 Compliance considerations (indicative; counsel required)

> The blueprint's legal and market statements came from secondary summaries and are **indicative only**. They must be reviewed by counsel before any public launch (open decision F7; question list in `sml/docs/founder/COUNSEL_QUESTIONS.md`).

Area	Design consequence
SEBI adviser rules	Education-only output; no buy/sell calls; router flags advice-seeking; the advisory note says the assistant is not an adviser. Whether this is sufficient is a counsel question
DPDP Act and Rules	Notice generated from the `FLOWS` registry; consent records on device; deletion path; retention rule (a 90-day draft exists); business entity and grievance officer not yet defined (`ENTITY` is null on purpose)
Data minimisation	Counters only; redacted, capped, local samples; no raw message collection without consent
Records of evidence	Tracker, manifests and model cards are the audit trail of how a model was judged

12.7 Incident and governance



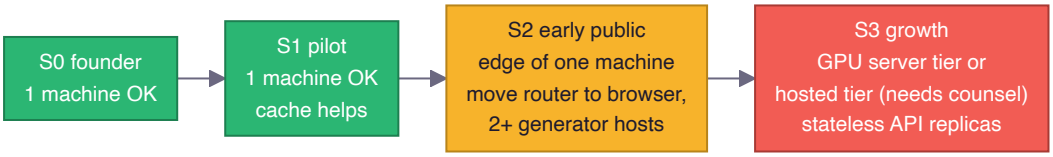
13. Performance, Scalability and Reliability (Phase 10)

13.1 Workload assumptions come first

No production workload exists. The founder's machine is the only user. Targets below are therefore **PROPOSED** and derived from measured per-request costs plus explicit scenario assumptions (A2). Replace them with real traffic once the shadow week and a pilot produce data.

Scenario	Users	Assumed behaviour (ASSUMPTION)	Peak answers per hour	Peak concurrent
S0 Founder	1	Sequential, interactive	20-60	1
S1 Pilot	50	5 questions per user per day, 20% in the busiest hour	50	1-3
S2 Early public	1,000	5 per user per day, 15% in the busiest hour	750	5-10
S3 Growth	20,000	same	15,000	100+

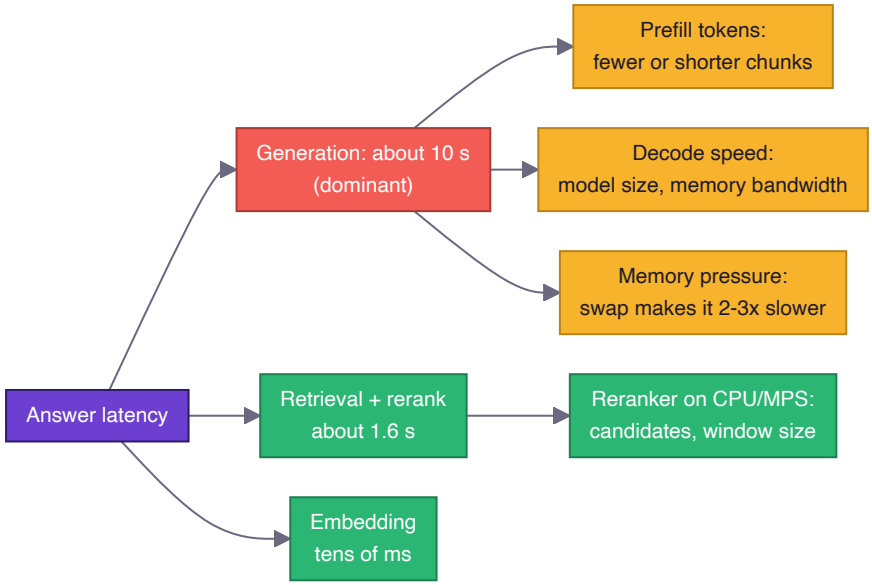
Capacity arithmetic. One Ollama instance with `NUM_PARALLEL=2` and a median answer time of about 10 s serves roughly 2 / 10 = 0.2 answers per second, about 720 per hour at best (measured per-request cost; real throughput on a swapping 24 GB Mac is lower, 2-3x slower under memory pressure). That covers S0 and S1 comfortably, S2 only at the edge, and S3 not at all without a different topology.



13.2 Measured cost per stage (CONFIRMED where measured)

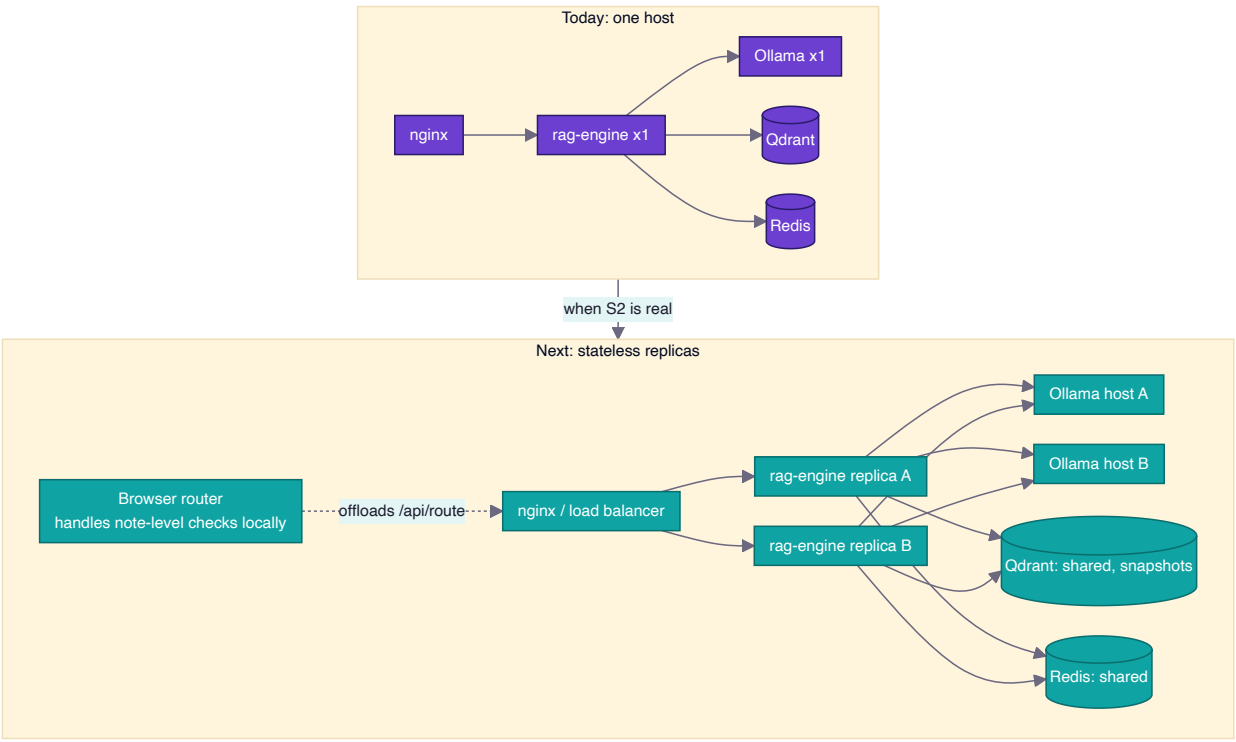
Stage	Cost	Source
Router decision (server)	about 33 ms (embedding dominates); load 0.8 s once	TRACKER H3
Router round trip from the browser	56-122 ms	TRACKER C9
Browser TF-IDF router	0.022 ms per message warm; model load + index 16 ms	TRACKER J1
Retrieval incl. personal quota	1.6 s median (was 1.9 s)	TRACKER H15
Generation	about 10 s median per answer (qwen3:8b, about 20 tok/s); qwen3:14b 22-28 s	TRACKER B12
Prefill	about 200-260 tokens/s, the dominant cost, so fewer/shorter chunks cut latency	`server.py` comments
HyDE	+4-7 s, no ranking gain; off	`QueryRequest`
Multi-hop	+2-5 s; off	`QueryRequest`
Full public benchmark	112 questions, roughly 20-25 minutes	PLAN

13.3 Bottleneck analysis



Bottleneck	Lever	Measured effect
Generation decode	Smaller or faster model; GPU host	14b: same score, 2.5x slower; 8b chosen
Prefill	Chunks 3 x 500 chars (2x500 and 3x300 both lowered quality)	Defaults retained
Memory swap	Free RAM, stop other VMs; `KEEP_ALIVE=30m`	Queries 2-3x slower under pressure
Repeated questions	Redis answer cache (1 h)	Hit rate not yet measured
Router load	Move the note-level router into the browser	0.022 ms vs 56-122 ms

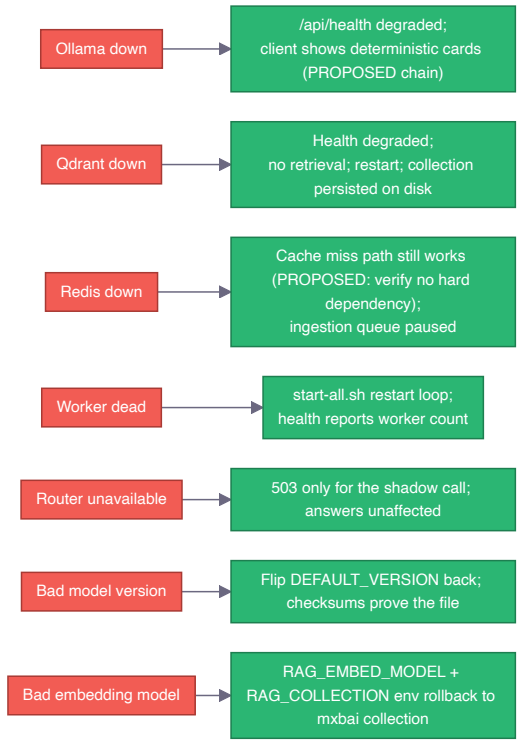
13.4 Scaling architecture (PROPOSED path)



Dimension	Approach	Condition
Vertical	More unified memory or a GPU host for generation	First step; no code change
Horizontal	`rag-engine` is stateless apart from Redis and Qdrant, so replicas behind nginx work; per-process state to check: in-process auth cache and model singletons	S2
Load balancing	nginx `upstream` with least-connections (LLM calls are long)	Replicas exist
Concurrency	`NUM_PARALLEL` per Ollama host; one RQ worker per source type	Measure first
Parallel processing	Ingestion jobs in RQ; benchmarks run **one at a time** (memory)	Existing rule

Dimension	Approach	Condition
Caching	Redis answer cache with `pipeline_signature()` in the key; consider caching public-only answers more aggressively	Hit rate measured first
Database optimisation	Qdrant payload index on `namespace` and `doc_type`; FTS5 file on fast disk	Check index presence (PROPOSED)

13.5 Fault tolerance, redundancy, disaster recovery



Objective	Today	Proposed target
Availability (SML answers)	Best effort on one machine	99% monthly for pilot (ASSUMPTION), with deterministic fallback so the app itself stays up
RPO (data loss tolerance)	Public index can be rebuilt from sources; user data is on device	Daily Qdrant snapshot; user data unaffected
RTO (recovery time)	Manual `start-all.sh`, minutes	Under 30 minutes with a documented runbook
Redundancy	None	Second Ollama host only at S2

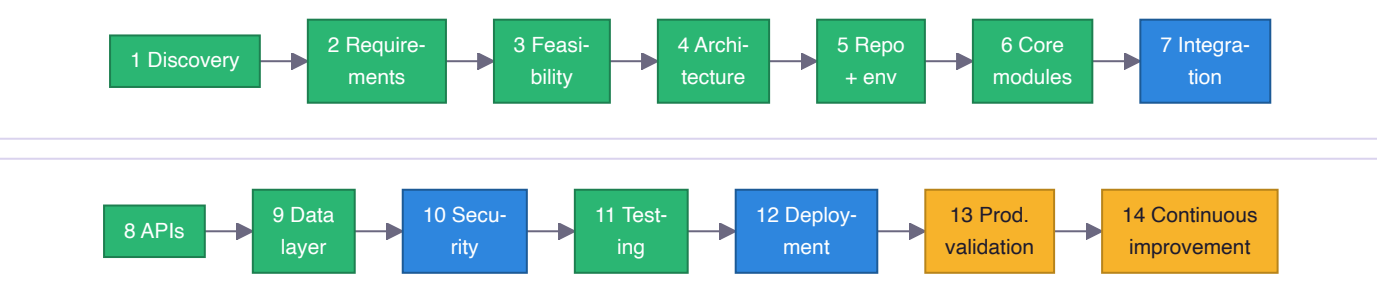
13.6 Performance testing strategy

Test	Purpose	Tool	Pass criterion (PROPOSED)
Quality benchmark	Detect quality regressions	`benchmark.py` (112), `benchmark_personal.py` (41)	Public ≥ 95 now, ≥ 100 target; personalised ≥ 38; isolation 5/5
Golden recall	Retrieval quality independent of generation	golden harness	recall@5 no lower than baseline minus 2 points
Latency profile	Per-stage timing	Prometheus histograms, benchmark timings	p50 ≤ 12 s, p95 ≤ 30 s at S1
Load test	Concurrency behaviour	k6 or Locust against `/api/query` (PROPOSED)	No errors up to the S1 peak; graceful 429 beyond
Soak	Memory growth, swap	4-hour mixed run	No monotonic RSS growth
Browser budget	Page weight	`tests/perf_budget.py`	Critical JS budget respected

14. Development and Implementation Strategy (Phase 11)

14.1 The fourteen stages

Status shows what has already happened for the SML. The pattern is **vertical slices behind flags**, never a model in isolation.



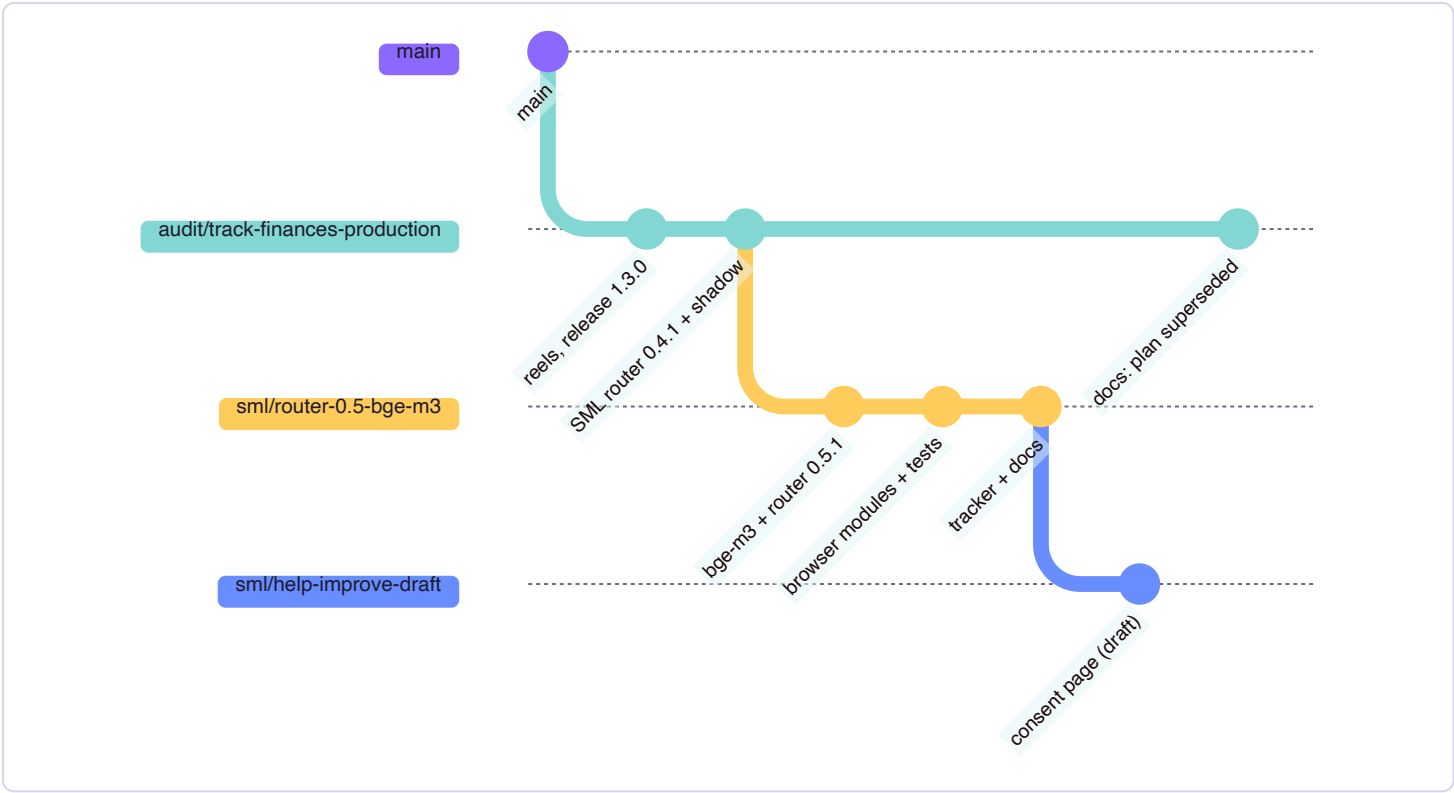
#	Stage	Objectives	Key tasks	Deliverables	Depends on	Skills	Tools	Completion criteria	Risks	SML status
1	Product discovery	Define what SML is and is not	Blueprint, non-goals, personas	Blueprint (14 Oct 2026)	none	Product, ML	Docs	Scope signed off	Scope creep	DONE
2	Requirement analysis	Testable requirements	FR/NFR/PR with IDs and gates	Section 4-5 of this document	1	Analyst	Tables	Every requirement has a verifier	Untestable wording	DONE (this document)
3	Technical feasibility	Prove the ladder works	Router baseline vs regex; retrieval triage; verifier study	Tracker evidence, honest negatives	2	ML	Python, benchmarks	Go/no-go per component	Synthetic results that do not transfer (verifier)	DONE
4	Architecture design	Layers, components, flows	Sections 6-9	This blueprint	3	Architect	Mermaid	Diagrams reconcile with code	Drift	DONE
5	Repo and environment	Reproducible dev	`sml/` folder rule, `verify.sh`, CI	Repo layout (Section 15)	4	DevOps	Git, CI	Fresh clone runs tests	Untracked artefacts	DONE
6	Core module development	Router, retrieval, checker	Train versions, rerank window, checker	Code + model cards	5	ML, JS, Python	scikit-learn, FastAPI	Gates measured	Overfit to agent-written text	PARTLY
7	Component integration	Wire behind flags	Shadow, advisory, client, report	Browser modules	6	JS	node tests	Flag off = no change	Cache-staleness after deploy	PARTLY
8	API development	Contracts	`/api/route`, `/api/query`	Endpoints + tests	6	Backend	FastAPI	Contract tests pass	Mixed error shapes	DONE (error contract PROPOSED)
9	Data layer	Stores and pipelines	Ingestion, namespaces, datasets	Qdrant, FTS5, Redis, manifests	6	Data eng	Qdrant, Redis	Isolation 5/5; no leakage	Source breakage	DONE
10	Security implementation	Controls from Section 12	Auth, flows registry, PII, loopback	Tests + notice	8, 9	Security	Presidio	Privacy tests 100%	Open: TLS, secrets manager, audit log	PARTLY
11	Testing	Section 16	Unit to sealed-set	Test suites	6-10	QA	pytest, node, Playwright	Gates green	Skipped tests masking gaps	DONE (ongoing)
12	Deployment	Section 17	Compose, Vercel, CI	Pipelines	11	DevOps	Docker, Actions	One-command release	Hosting decision D1 open	PARTLY
13	Production validation	Real behaviour	Shadow week, report, review	Report with intervals	12	Founder, reviewer	`smlReport()`	≥ 300 messages, reviewed	Counters hold no text	NOT STARTED (needs founder)
14	Continuous improvement	Loop	Retrain, sealed sets, roadmap	New versions	13	All	Tracker	Each loop passes its gate	Metric gaming	ONGOING

14.2 Coding standards and working rules

Area	Standard	Source
Module pattern	UMD so a file works in the browser and in node tests	`js/sml/*`
Fail behaviour	Safety and counter modules fail open and never throw into the page	design rule
Flags	Every user-visible SML feature is behind a localStorage flag, default off	design rule
Config	Tunables are env knobs in `config.py`, range-clamped, and part of `pipeline_signature()`	H5, A10
Models	Version required to train; no overwrite without `--overwrite`; checksums; model card per version	H4, F9
Data	Held-out sets never trained or tuned on; sealed sets scored once	principle 4
Network	Any new host must appear in `FinosPrivacy.FLOWS` in the same change	principle 5
Guards	Add guards, never remove `AryaGuardrails`	principle 6
Commits	The founder commits; commit messages carry no AI attribution	founder rules

14.3 Branching and release strategy

Current practice [CONFIRMED]: long-lived feature branches (`audit/track-finances-production`, `sml/router-0.5-bge-m3`, `sml/help-improve-draft`) with grouped commits and pull requests against the branch they build on.



PROPOSED simplification: trunk-based development with short-lived branches, every SML change flagged default-off, and a merge only when `verify.sh` is green. Model and data changes get their own commits (so a rollback is one revert).

14.4 Documentation standards

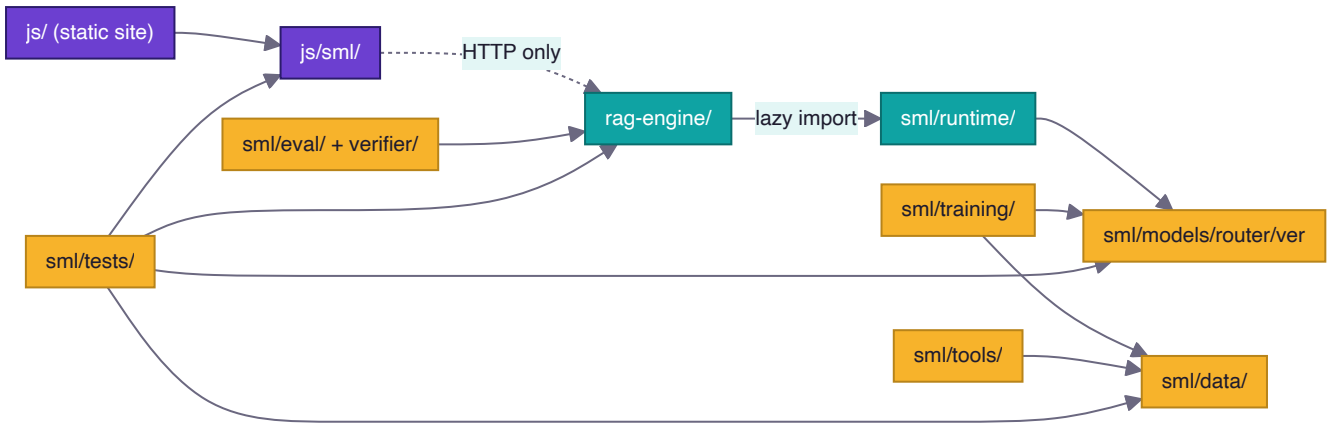
Document	Rule
`TRACKER.md`	A row with status and evidence after **every** step
`PLAN.md / ROADMAP.md`	Updated in the same change as the work
Model card	Per version: data, settings, sealed results, limits, files, digests
Dataset manifest	SHA-256 of every file; leakage tests discover files automatically
This blueprint	Architecture decision records (Section 23) for any change to a decision

15. Repository and Code Organization

15.1 Repository layout (CONFIRMED)

```
Initial Deployment/
├── html/ , js/ , css/ , calculators/ , sw.js , manifest.json     static PWA (Vercel serves the root)
├── js/sml/                                                        browser SML modules (must stay under js/)
│     normalizer.js  numeric-check.js  router-shadow.js  router-advisory.js
│     router-client.js  tfidf-router.js  pulse-counters.js  report.js
│     sample-capture.js  categoriser-export.js
├── rag-engine/                                                    FastAPI service :7476
│     server.py  config.py  jobs.py  scheduler.py  metrics.py  schema.sql
│     ingestion/  retrieval/  generation/  embedding/  storage/  evaluation/
├── sml/                                                           everything SML except browser JS
│     PLAN.md  ROADMAP.md  TRACKER.md  README.md
│     data/  training/  runtime/  models/router/<ver>/  eval/ (golden/)
│     verifier/  categoriser/  rules/  tools/  scripts/  tests/ (js/)  docs/
├── arya-ai/ , alerts/ , stock-engine/ , document-ai/ , voiceagent/ , gateway/   other back-ends
├── docker-compose.yml , nginx-gateway.conf                        deployment
├── supabase/ (SQL)                                                account schema and RLS
├── tests/                                                         site-wide JS + Python + Playwright tests
├── mobile/                                                        Expo companion app
└── .github/workflows/ci.yml                                       CI
```

15.2 Module organisation and dependency direction



Rules. The browser never imports server code; `rag-engine` imports `sml/runtime` lazily; training code never ships to production; tools default to local files and send nothing anywhere.

15.3 Configuration and environment management

Layer	Mechanism	Example
Server tunables	Environment variables with clamped defaults	`RAG_PROMPT_TOP_K`, `RAG_EMBED_MODEL`, `RAG_COLLECTION`, `RAG_PERSONAL_QUOTA`
Browser features	localStorage flags, default off	`finos_sml_router_shadow`, `finos_sml_router_advisory`, `finos_sml_router_local`
Secrets	`.env` per service (not in repo)	`SUPABASE_*`
Cache safety	`config.pipeline_signature()` in the answer-cache key	prevents stale answers
Model version	`DEFAULT_VERSION` in `router_predict.py`	`0.4.1`

16. Testing and Quality Assurance (Phase 12)

16.1 Testing pyramid for a statistical system

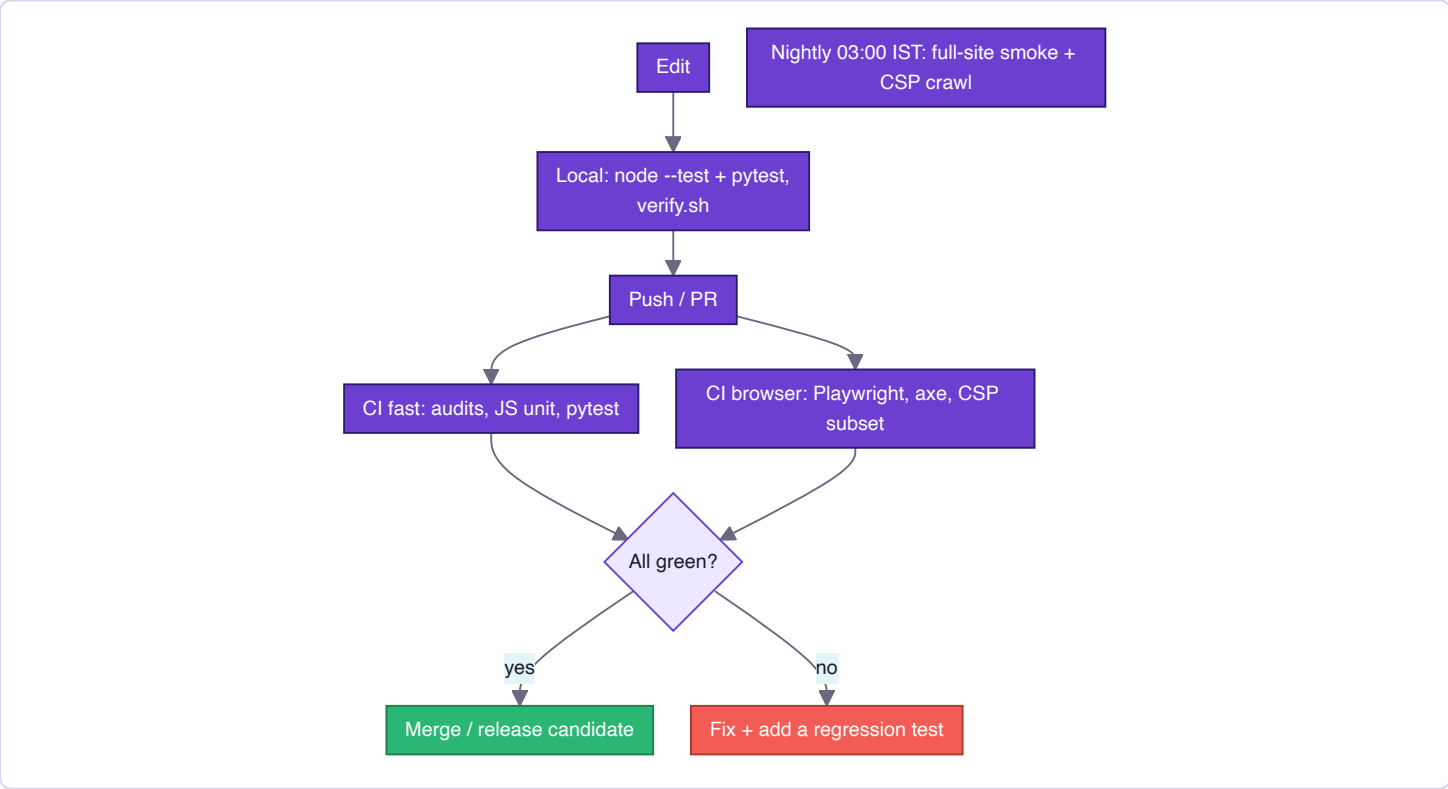
The classical pyramid has a new top: tests on **sealed data** that measure rates, not equality.



16.2 Test categories, purpose and where they apply

Category	Purpose	Applies to	Existing examples
Unit	Logic in isolation	number checker, counters, normaliser, report	`sml-numeric.test.js` , `sml-router-shadow.test.js`
Integration	Components together	endpoints, namespaces, personal quota, route endpoint	`test_namespace_isolation.py` , `test_route_endpoint.py` , `test_personal_quota.py`
System / E2E	Whole path in a real browser	Arya panel, shadow mode, PWA	Playwright suites, manual real-browser runs recorded in the tracker
Regression	Nothing got worse	public/personalised benchmarks, sealed sets after retrain	`benchmark.py` , `eval_fresh.py`
Data / leakage	Evidence integrity	all datasets	`test_router_data.py` : exact and near-duplicate overlap = failure
Parity	Two implementations agree	Python vs JS TF-IDF router	`sml-tfidf-router.test.js` (320 texts, 0.0003 max difference)
Reproducibility	A retrain yields the same model	router	retrain as 9.9.9: identical `eval.json` , coefficient difference 0.0
Integrity	Files unchanged	model artefacts	`SHA256SUMS` , `verify.sh`
Performance	Speed and budgets	latency, page weight	benchmark timings, `perf_budget.py`
Load / stress	Concurrency limits	`/api/query`	PROPOSED (k6 or Locust)
Security	Abuse resistance	namespaces, flows, redaction	`privacy.test.js` , dirty-file validator tests, ReDoS probes
Compatibility	Browsers and devices	PWA, router JS	Playwright Chromium; device lab PROPOSED
Failure	Behaviour under faults	fail-open modules, 503/422 contracts	module tests that throw; endpoint tests
Accessibility	WCAG checks	pages	axe in CI
UAT	Real people, real tasks	label review, shadow week	founder pack, review pages

16.3 Test execution lifecycle



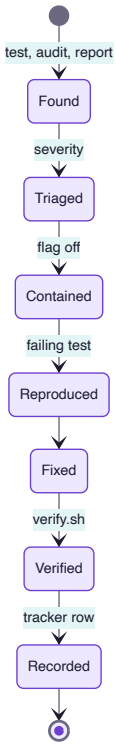
16.4 Quality gates

Gate	Condition	Where defined
R (router as active guard)	macro-F1 ≥ 0.95, recall lower bound ≥ 0.98 with ≥ 300 risky, false alarm per level, sealed set, reviewed real data	ROADMAP 3.3
B (retrieval)	public ≥ 100 / 112 twice, personalised ≥ 38 / 41, isolation 5/5, golden recall@5	ROADMAP 3.3
J (browser router)	decision ≤ 100 ms, parity, size ≤ 1 MB gzip, false alarm ≤ 5%	ROADMAP 3.3
V (verifier)	P ≥ 0.90 and R ≥ 0.80 on a founder-labelled real set	ROADMAP 3.3 (NO-GO today)
C (categoriser)	accuracy ≥ 0.90 at coverage ≥ 85% on a merchant-disjoint set ≥ 300	ROADMAP 3.3
P (ranker)	replay uplift interval excludes zero on ≥ 1,000 impressions	ROADMAP 3.3
E (embedding)	+3 points on the Hinglish/Hindi slice, interval excludes zero, no English regression	ROADMAP 3.3
Always	`verify.sh` green; no undeclared host; held-out sets untouched	CI + tests

16.5 Test coverage strategy

Coverage is judged by **risk**, not line percentage: every requirement in Section 5.2 has a named verifier; every bug found gets a regression test (the audit found 9 bugs and added tests for each); every held-out data file is covered automatically by the manifest test; skipped tests are counted and explained (6 JS and 4 Python skips in a clean snapshot, caused by absent model files or services, not by failures).

16.6 Defect management



16.7 Release acceptance criteria

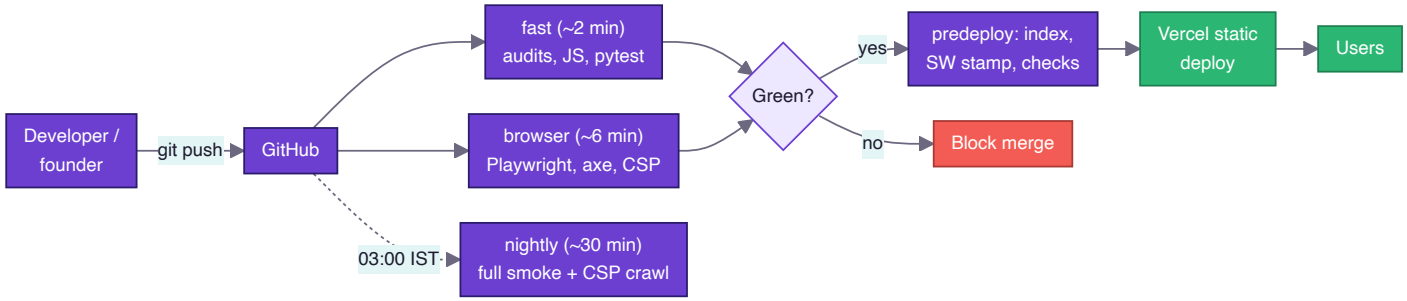
Release type	Must hold
Code only	`verify.sh` green; flag default unchanged; browser check if UI touched
Retrieval change	Public ≥ 95 and personalised ≥ 38 with isolation 5/5 against a same-day baseline; cache signature updated
Router version	Card + `eval.json` + `SHA256SUMS`; settings chosen on dev only; one sealed-set score; leakage tests green; predictor default changed only by explicit decision
Any new network flow	`FLOWS` entry and a passing host-coverage test

17. Deployment and DevOps (Phase 13)

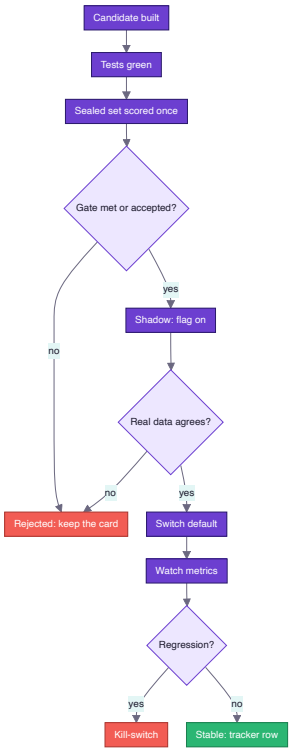
17.1 Environments

Environment	What it is today	Purpose
Development	Founder Mac: `start-all.sh` starts Qdrant, the server and an RQ worker; Ollama separately; static site on a local HTTP server	Build and measure
CI	GitHub Actions runners	Tests, audits, browser suites
Staging	Vercel preview deployments for the static site (CORS allow-list already accepts `finos1-*.vercel.app` previews); no staging for the Python engine	Check UI changes
Production	Vercel for the static site; **engine hosting undecided** (decision D1)	Serve users

17.2 CI/CD architecture



17.3 Release lifecycle for a model or retrieval change



17.4 Rollbacks (CONFIRMED mechanisms)

Change	Rollback
Router version	Change `DEFAULT_VERSION`, restart; every version's files and checksums are kept
Browser feature	Remove the localStorage flag; modules fail open
Retrieval knob	Unset the env variable (defaults equal today's behaviour); cache signature changes so no stale answers
Embedding model	Start with `RAG_EMBED_MODEL=mxbai-embed-large RAG_COLLECTION=finos_chunks` (old collection kept; note content ingested after the switch exists only in the new collection)
Static deploy	Vercel redeploy of the previous commit; service worker version stamp refreshes caches

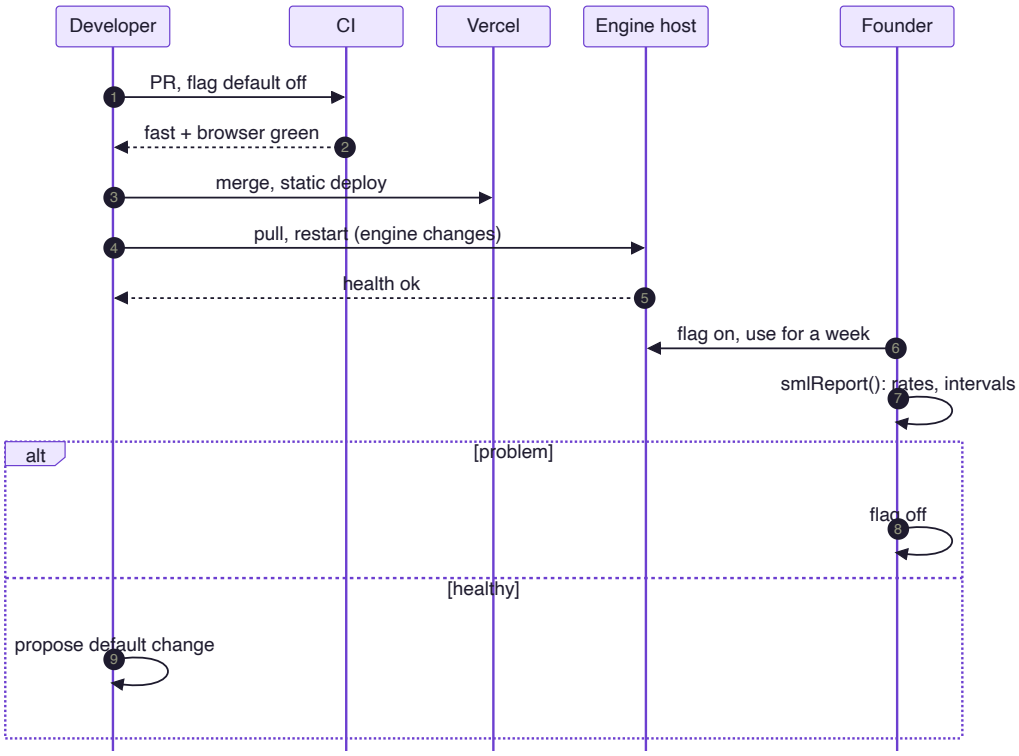
17.5 Containers and infrastructure as code

Item	State	Note
Docker Compose	In use: Ollama (loopback port), alert-engine, voice-agent, stock-engine, document-ai, market-intel, arya-ai, rag-engine, gateway-health, nginx gateway on :8000	Health checks on services; `restart: unless-stopped`
Dockerfiles	Per service (`rag-engine/Dockerfile` etc.)	
Infrastructure as code	Compose + nginx config under version control	Terraform or Pulumi **not justified** until a cloud host is chosen
Environment variables	`.env` per service; tunables as env knobs	Add `.env.example` per service (PROPOSED)

17.6 Logging, metrics, tracing, alerting

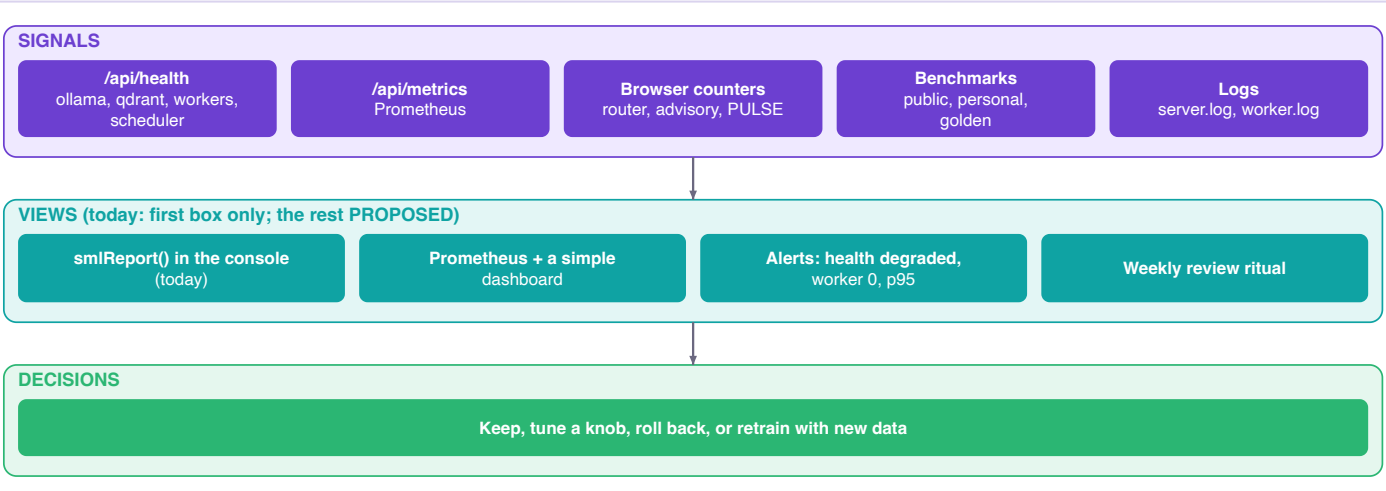
Signal	Today	Proposed
Logs	`server.log`, `worker.log` files	Structured JSON logs, request-id field, rotation; **never message text**
Metrics	Prometheus: request latency histogram, cache hits/misses, ingestion results, faithfulness flags, feedback votes, router decisions	Add per-stage latency, queue depth, Ollama time-to-first-token
Tracing	None	Request-id correlation first; OpenTelemetry only if multi-service latency becomes unclear
Alerting	None	Alert on health `degraded`, worker count 0, p95 latency, error rate

17.7 Production release lifecycle



18. Monitoring and Maintenance (Phase 14)

18.1 What is observed, by whom



Category	Mechanism today	Gap
Application monitoring	Prometheus counters and histograms; `/api/health`	No dashboard
Infrastructure monitoring	Health shows Ollama, Qdrant, worker count, next scheduled runs; gateway aggregates all services	No host CPU / memory / swap history (swap was about 25 GB at times)

Category	Mechanism today	Gap
Error tracking	Error counters (`errors` in shadow stats; ingestion result counters)	No error aggregation service
Performance monitoring	Request latency histogram by endpoint; benchmark timings	No per-stage breakdown
Resource monitoring	Manual (`top`)	Add host metrics exporter
Audit trail	Tracker, manifests, model cards (evidence audit)	No access audit log for personal data
Log management	Plain files	Rotation, structure, no message text
Incident management	Section 12.7 loop	A written runbook (PROPOSED)
Root-cause analysis	Tracker rows with reproduction (examples: the 38 → 35 personalised slip proven to be corpus growth by re-running the old code)	Template (PROPOSED)
Version management	Pipeline version, router version, embedding model + collection, model cards	A single "release manifest" file (PROPOSED)
Dependency updates	Pinned requirements; CI installs fresh	Scheduled review (PROPOSED), noting prometheus-client and sentence-transformers behaviour changes already bit once
Technical debt	Listed in `docs/PENDING.md` and the tracker's open items	A debt register with owners
Continuous optimisation	Pre-registered experiments with adopt rules	Keep: a dropped lever is a valid result

18.2 SLIs and SLOs (PROPOSED; ratify after a pilot)

SLI	Definition	SLO
Answer availability	Share of `/api/query` calls returning a complete answer	≥ 99% (pilot)
Answer latency	p50 / p95 of `/api/query` end-to-end	≤ 12 s / ≤ 30 s at S1
Router latency (server)	p95 of `/api/route`	≤ 150 ms
Router error rate	`errors / total` in shadow stats	≤ 2%
Ingestion freshness	Age of newest news chunk; days since last regulator ingest	News ≤ 4 h; regulators ≤ 8 days
Retrieval quality	Public benchmark, golden recall@5	≥ 95 / 112 and recall@5 ≥ baseline - 2 points
Isolation	Cross-user leak checks	100% (5/5)
Undeclared network calls	`privacy.test.js`	0
Fail-open integrity	Answers shown when a safety module throws	100%

18.3 Operational KPIs (product-facing)

KPI	Meaning	Today
Router guard rate with 95% interval	How often the router flags	Collected in the shadow week; 22 messages gave 9.1% (2.5%-27.8%), too few to judge
Router vs regex disagreement	Where the new layer adds or subtracts	Counters exist (`router_only` , `regex_only`)
Advisory fire rate on benign traffic	Annoyance risk	0.8% (0.5.1) to 2.2% (0.4.1) on held-out sets
Cache hit rate	Cost and latency saved	Counter exists; not yet reviewed
Facts coverage	Share of personal questions answered without a "gap"	PROPOSED
Label agreement (kappa)	Quality of evidence	0.77 for the verifier claim set; router review pending

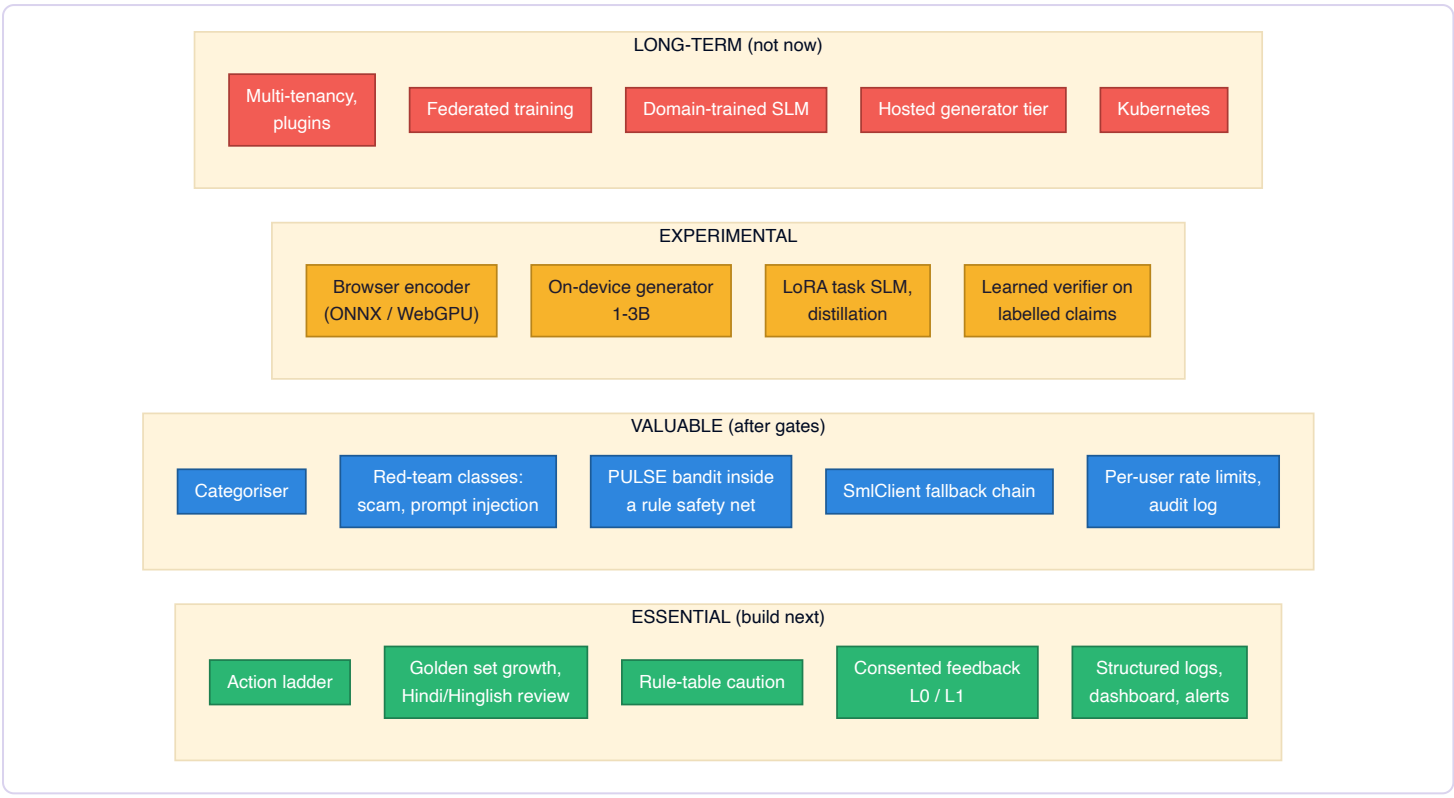
18.4 Weekly review ritual (PROPOSED)

1. Run `await AryaSidebar.smlReport()` ; note total messages, guard rate with interval, disagreements.
2. Run the golden recall and the public benchmark once each if anything changed.
3. Read open tracker rows; promote or drop each.
4. Decide one action: keep, tune a knob, retrain with new data, or roll back.

19. Advanced Capabilities (Phase 15)

> **Rule.** AI, distributed computing and complex infrastructure are introduced only when a measured problem needs them. Each item below states the trigger that would justify it. Items marked "not justified now" stay out of the design.

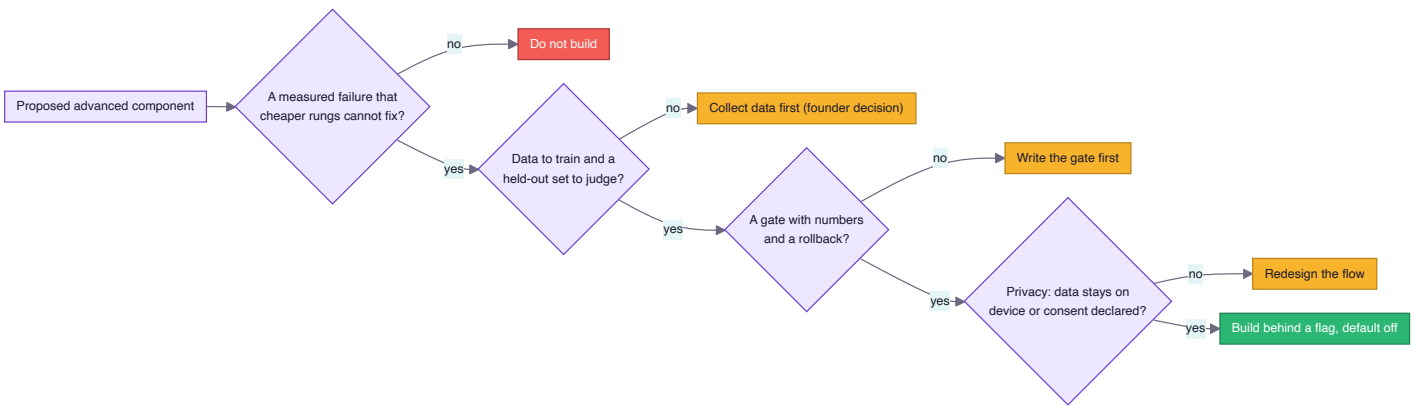
19.1 Classification



19.2 Evaluation of each possibility

Capability	Tier	Justified use case in FINOS	Trigger to start	Why not earlier
Intelligent automation (action ladder)	Essential	Real advice/evasion requests need a proportionate response	Founder answers F3; real shadow data reviewed	Needs an operating-point decision
Predictive capabilities	Valuable	Spend forecast, anomaly flags (blueprint use cases)	Interaction or statement data exists	No data; none is a trust item
Advanced analytics	Valuable	Offline replay of PULSE ranking	≥ 1,000 impressions	Counters are aggregates today
Real-time processing	Not needed	Voice already uses WebSockets; SML needs no streaming beyond SSE	n/a	No requirement
Event-driven systems	Already minimal	Ingestion jobs through RQ	n/a	A broker like Kafka has no use case
Distributed architecture	Long-term	Multiple generator hosts at S2+	Measured p95 > SLO at pilot scale	One machine suffices at S0-S1
Self-healing	Partly present	Worker restart loop exists	Repeated manual restarts logged	Keep simple
Multi-tenancy	Long-term	B2B / partner offering (blueprint: partner channels)	A signed partner	Single-tenant namespaces already isolate users
Extensible plugin architecture	Long-term	New doc sources, intents	Third parties asking to extend	Loaders per source already modular
Advanced governance	Valuable	Model registry manifest, access audit	Hosted deployment	Single operator today
Intelligent optimisation	Experimental	Automatic threshold tuning	Real data and a safe tuning set	Violates "no tuning on test sets" unless a proper dev stream exists

19.3 Decision tree for adding any AI or distributed component



20. End-to-End Engineering Roadmap

20.1 Seven stages

Estimated complexity: **S** (days), **M** (1-2 weeks), **L** (several weeks), **XL** (a quarter). Estimates are planning guesses, not measurements. "Done" marks what the repository already delivers.

Stage 1: Foundation (DONE)

Item	Detail
Tasks	Define SML, non-goals, principles; personas; requirements; scope
Deliverables	Blueprint (14 Oct 2026), this document's Sections 1-5
Prerequisites	Product vision
Dependencies	none
Completion criteria	Every requirement has a verifier and an owner component
Complexity	M (done)
Risks	Ambiguity of "SML" (assumption A1)

Stage 2: Architecture (DONE for the current scope)

Item	Detail
Tasks	Layers, components, data stores, technology selection, decisions
Deliverables	Sections 6-10, 23
Prerequisites	Stage 1
Dependencies	Hosting decision D1 for production topology
Completion criteria	Diagrams reconcile with code; decision records written
Complexity	M
Risks	Drift between diagrams and code (mitigation: re-verify on each release)

Stage 3: Design (PARTLY)

Item	Detail
Tasks	Standardise the error contract; action-ladder specification; manifest for model delivery; DB schema application check; consent text approval
Deliverables	API error envelope; ladder spec with per-level thresholds; `manifest.json` schema; counsel-reviewed consent text
Prerequisites	F3, F7 answers
Dependencies	Stage 2
Completion criteria	Specs reviewed; thresholds justified from sealed-set curves
Complexity	M
Risks	Waiting on people (founder, counsel, reviewer)

Stage 4: Development

Item	Detail
Tasks	Wire the action ladder behind a flag; head tuned for a false-alarm target for the browser; static model delivery with hashes (J2); golden set growth and Hindi review; rule-table caution; categoriser when rows arrive; red-team classes; structured logging and alerts
Deliverables	New modules and versions with model cards; tests
Prerequisites	Stage 3 decisions; data (rows, reviewers)
Dependencies	`verify.sh` green on the branch
Completion criteria	Each item meets its gate on a sealed or golden set
Complexity	L
Risks	Overfitting to agent-written text; label noise on boundary cases

Stage 5: Validation

Item	Detail
Tasks	Shadow week and review; second-reviewer labels (kappa); sealed sets for each new version; load test; red-team set; accessibility and browser matrix; security review of flows
Deliverables	Report with intervals; new sealed-set result files; test reports
Prerequisites	Stage 4 candidates; real data
Completion criteria	Gate R/B/J/C as relevant, or a written no-go
Complexity	L
Risks	Small real sample; gate missed by a hair (as 0.5.1 missed recall bound by 0.006)

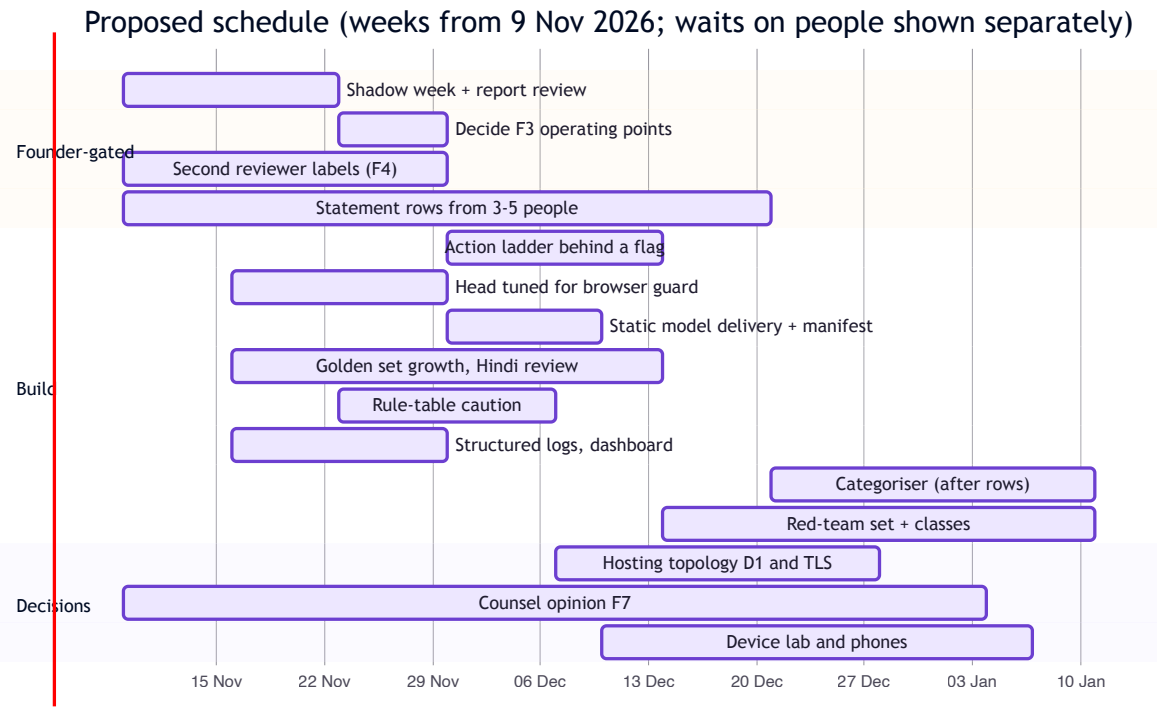
Stage 6: Deployment

Item	Detail
Tasks	Choose hosting topology (D1); TLS; secrets; backups; staging for the engine; release manifest; runbook; alerts
Deliverables	Hosted engine or browser-only note-level guard; IaC (compose + env templates); runbook
Prerequisites	D1, counsel opinion F7, domain and TLS
Completion criteria	One-command release and rollback rehearsed; health checks green; SLOs defined
Complexity	L
Risks	Hosting a local-first design; cost; legal exposure

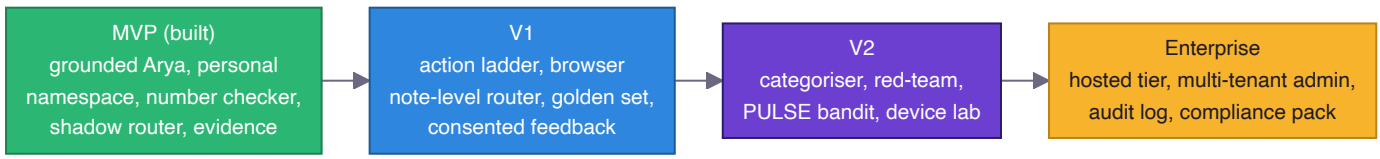
Stage 7: Optimisation

Item	Detail
Tasks	Monitoring and weekly ritual; cache tuning; bandit ranker if usage exists; device lab; scaling per Section 13; dependency updates; debt register
Deliverables	Dashboards, quarterly re-baseline of models and golden set
Prerequisites	Stage 6 in production
Completion criteria	SLOs met for a month; roadmap re-baselined
Complexity	XL (ongoing)
Risks	Metric gaming, model churn

20.2 Dependency-aware schedule (PROPOSED, starting week of 9 November 2026)



21. MVP, V1, V2 and Enterprise Evolution



Release	Scope	Entry criteria	Exit gate	Main risk
MVP	As built (Section 1.2)	Tests green; isolation 5/5	Public ≥ 95 / 112 (met)	Founder-only usage
V1	Safe-by-default guard layer and measurable retrieval	F3 answered; shadow data reviewed	Gate R, Gate B, Gate J (note level)	False alarms annoy users
V2	Personal intelligence	≥ 1,000 statement rows; ≥ 1,000 impressions	Gate C, safety gate, Gate P	Data volume and diversity
Enterprise	Operate for others at scale	Counsel opinion; hosting decided; entity defined	SLOs met for a quarter; compliance review	Regulatory and ops burden

22. Technical Risks and Mitigation

ID	Risk	Likelihood	Impact	Mitigation	Early warning
R1	Router trained on agent-written text does not generalise to real messages	High	High	Shadow week, device-local samples, second reviewer, real data before any active mode	Disagreement rate far from sealed-set estimate
R2	Gate cannot be met at one threshold (recall vs false alarm)	Certain (seen twice)	Medium	Different operating points per action level; more targeted data	Sealed-set curves
R3	Generation errors (number from a neighbouring sentence) remain	High	High	Number checker, figures table, citations; no automated verifier	Benchmark class-A failures
R4	Small-sample statistics misread	Medium	Medium	Wilson/Clopper-Pearson intervals; 300-message readiness gate	Report shows wide intervals
R5	Test-set contamination	Medium	High	Leakage tests; manifests; sealed sets scored once	A test failing on a new file
R6	Engine hosted nowhere for users	High	High	Browser note-level router; decision D1; hosted tier only after counsel	No visitors protected
R7	Embedding switch breaks consumers	Medium	Medium	Router embeds with its own model; collection and model in cache key; rollback env vars	Retrieval or router errors after a model change
R8	Memory pressure on a 24 GB machine slows everything 2-3x	High	Medium	One heavy job at a time; free memory (F8); Ollama keep-alive settings	Swap usage
R9	Supabase project unavailable	Medium	Medium	SML does not depend on sync; monitor	Auth check failures
R10	Regulatory exposure from advice-like output	Medium	High	Education-only design, guardrails + router advisory; counsel (F7)	Advice-seeking guard rate rising
R11	Stale or wrong rules in the UI	Medium	Medium	Figures table; CA review of two flagged notes	Consistency check findings
R12	Prompt injection through ingested content	Medium	Medium	Red-team set; class addition; allow-list of source hosts	Odd instructions in retrieved chunks
R13	Labels inconsistent across reviewers	Medium	Medium	Labelling guide; kappa reporting; adjudication tooling	Kappa below 0.7
R14	Documentation drifts from code	High	Medium	Generated status file; tracker rule; this blueprint's reconciliation check	A doc number that no longer matches the code
R15	Uncommitted or unmerged work lost	Medium	High	Commit grouped branches; verify the branch you read from	Missing files in the working tree

23. Architecture Decision Records

Status: **Accepted** = in the code and measured; **Proposed** = recommended, not built; **Open** = needs a decision by the founder.

ADR	Title	Context	Decision	Consequences	Status
ADR-01	A ladder of small components, not one model	A big custom model is costly, hard to evaluate, and does not fix hallucinated numbers	Climb rungs only on measured need (rules, classical ML, encoders, prompted SLM + RAG, LoRA, domain model)	Rungs 4-5 deliberately not built; qwen3:14b gave no gain over 8b	Accepted
ADR-02	Numbers come from code, never model text	LLMs invent figures	Tools and sources supply figures; a deterministic checker verifies; the checker never edits	Misattribution not caught; figures table needed	Accepted
ADR-03	Modular monolith, no new service in V1	One host, small team	Router inside `rag-engine`; browser router is a static file	Fewer moving parts; scaling path documented	Accepted
ADR-04	Ollama as the local model runtime	Local, private, simple	One daemon for generation and embeddings; loopback binding	No auth on Ollama: must never be exposed	Accepted
ADR-05	Hybrid retrieval: dense + BM25 with RRF, then rerank	Exact terms (80CCD(1B)) and meaning both matter	Qdrant + FTS5, k=60 fusion, bge-reranker on the query window	Two indexes to keep in step	Accepted
ADR-06	bge-m3 for retrieval; mxbai stays for the router	bge-m3 lifted draft Hindi/Hinglish recall@5 0.425 → 0.900 with a 1-point English dip	Adopted 26 Oct 2026; old collection kept as rollback; router embeds with its own model	Two embedding models installed; Hindi slice unreviewed	Accepted
ADR-07	Router = blend of TF-IDF and embedding logistic heads	Small, fast, exportable, explainable	13 intents, guard on class or risk mass	Weakest on Hindi and boundary labels	Accepted
ADR-08	Shadow mode first, counters only	A wrong guard hurts users; real text is sensitive	Count, never store text; flags default off	The review gap (no text) solved by opt-in device-local samples	Accepted
ADR-09	Sealed sets scored once; settings on dev only	Tuning on tests inflates results	Pre-registered rules; leakage tests; manifests	Needs fresh blind data each round	Accepted
ADR-10	No automated faithfulness gate	Synthetic P 0.997 collapsed to 0.17 on real answers; best judge combos missed the gate	Keep the numeric check; collect founder-labelled claims	Misattribution remains a risk	Accepted
ADR-11	REST + SSE; no GraphQL, no WebSocket for the SML	Fixed shapes, one-way stream	Keep contracts simple	None	Accepted
ADR-12	Docker Compose now; no Kubernetes	Single-host scale	Revisit at Enterprise	Manual scaling	Accepted
ADR-13	Every feature flagged default-off and fail-open	Safety and reversibility	localStorage flags; try/catch wrappers	No central kill-switch	Accepted

ADR	Title	Context	Decision	Consequences	Status
ADR-14	No pickle in shipped artefacts	Unsafe deserialisation	`allow_pickle=False` ; pure-numpy TF-IDF export tracked; joblib git-ignored	Retrain script must reproduce	Accepted
ADR-15	Namespace isolation enforced by a verified token	Personal data must not cross users	`user:{id}` namespace + token check per query	Auth depends on Supabase	Accepted
ADR-16	Education-only output policy	Regulatory exposure	Guardrails + advisory note + prompt rules	Counsel must confirm	Accepted (legal review Open)
ADR-17	Separate operating points per action level	Recall and false alarm cannot both hit gate at one threshold (v7, v8)	Note, steer and block each get their own threshold	Needs founder numbers	**Open (F3)**
ADR-18	Topology: on-device-first hybrid	Public visitors have no engine	Browser note-level router now, encoder and generator later	Needs reference devices	**Open (F5, D1)**
ADR-19	Standard error envelope and `/api/v1` only when breaking	Mixed error shapes	One envelope; version paths later	Small client change	Proposed
ADR-20	Structured logs, dashboard and alerts	No visibility beyond counters	JSON logs without message text; Prometheus + dashboard	Operational work	Proposed

24. Complete Glossary

Term	Meaning
Abstain	Returning "not sure" instead of a guess, usually below a confidence threshold
Action ladder	Proportionate responses to a flagged message: note, steer, block
Advisory (router advisory)	A one-line, appended note shown when the router sees advice-seeking that the regex missed
Arya	FIN-OS's AI assistant panel
BM25	A keyword ranking function; here SQLite FTS5 `bm25()`
Bandit (contextual)	An algorithm that chooses among options while learning from taps; planned for PULSE
bge-m3 / bge-reranker-v2-m3	A multilingual embedding model / a cross-encoder reranker
Bi-encoder / cross-encoder	Embeds texts separately for fast search / reads query and text together to score
Blind writer	An independent author given only category descriptions to write test or training messages
Chunk	A slice of a document with metadata, the unit of retrieval
Clopper-Pearson	An exact confidence interval for a proportion; used for recall lower bounds
Cohen's kappa	Agreement between two labellers corrected for chance
Counters	Numeric tallies kept in the browser; never message text
DPDP Act	India's Digital Personal Data Protection Act, 2023
Embedding	A numeric vector representing meaning
Fail-open	If a check cannot run, the answer is still shown
False alarm	A harmless message flagged as risky
FinosStore / FinosFacts / FinosTaxCore	The user data store / personal facts and gaps / shared tax computations
FLOWS	The registry of every data flow, which generates the privacy notice
Gate	A numeric pass condition on held-out data (R, B, J, V, C, P, E)
Golden set	Fixed, human-verified questions with gold passages, used to gate releases
Guardrails	Deterministic regex layer that refuses illegality and adds notes
HyDE	Hypothetical document embeddings: generate a fake answer to search with; off by default
Intent	One of 13 message categories the router outputs
LoRA / QLoRA	Low-rank adapters for fine-tuning a model cheaply
Macro-F1	Average F1 over classes, treating each class equally
Namespace	A retrieval partition: `public` or `user:{id}`
NLI	Natural language inference: does a source support a claim?
Numeric checker	The deterministic module verifying answer figures against evidence
Operating point	A chosen threshold on the recall-false alarm curve
Ollama	The local model server
PIPELINE_VERSION / pipeline_signature	Version and tunables string that keys the answer cache
PULSE	The 24-section personalised tab in Arya
Qdrant	The vector database
RAG	Retrieval-augmented generation
Recall@k	Share of questions whose gold passage appears in the top k results
Reranker	A second-pass scorer that reorders candidates
RQ	Redis Queue for background jobs
RRF	Reciprocal rank fusion: merges ranked lists with 1 / (k + rank)
Router	The classifier mapping a message to an intent and risk
Rung	A level of the model ladder

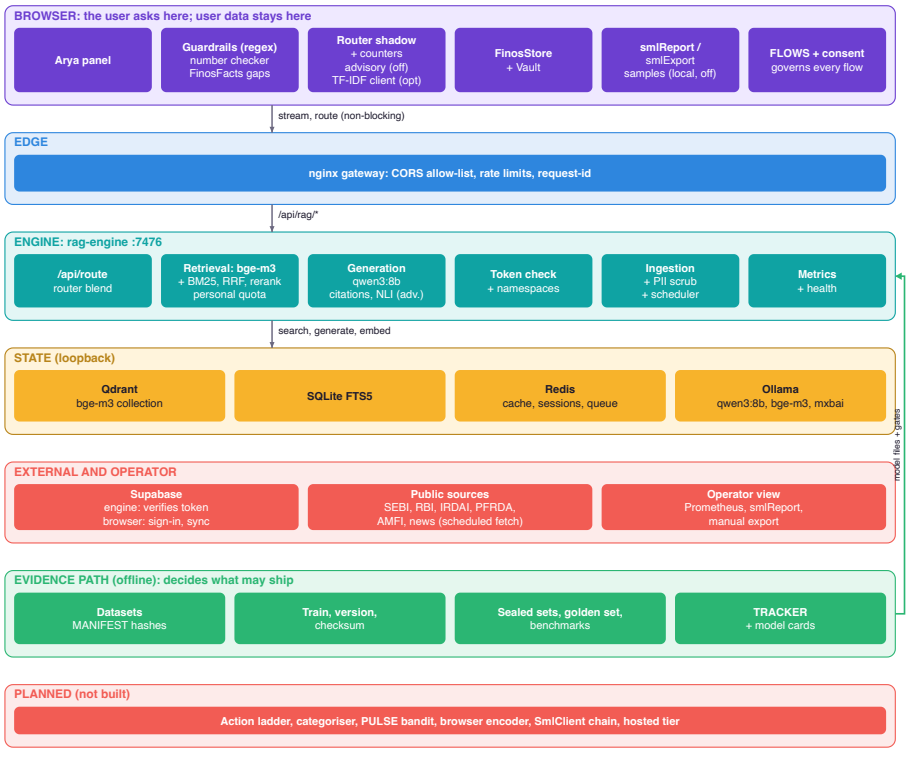
Term	Meaning
Sealed set	A test set written after a model is frozen and scored exactly once
Shadow mode	Observe and count without changing any answer
SLI / SLO	A measured service quality / its target
SML / SLM	Small model layer / small language model
SSE	Server-Sent Events: one-way streaming over HTTP
TF-IDF	Term frequency weighting used for the router's word and character features
Verifier	A model checking whether an answer is supported by sources (NO-GO as a gate)
Wilson interval	A confidence interval for a proportion that behaves well for small samples

25. Final System Architecture Summary

25.1 What the SML is, in six lines

1. A **ladder of small, measured components** around Arya, not a single model.
2. **Numbers come from code**, checked deterministically; the model explains, it does not compute.
3. **Personal data stays on the device** by default; every network flow is declared; telemetry is counters only.
4. A **router** (13 intents, 3 languages) is in shadow mode with a flag-gated soft advisory and an opt-in browser TF-IDF client; it is not yet an active guard.
5. **Retrieval** is hybrid (bge-m3 dense + BM25 + RRF + reranker + personal quota); qwen3:8b writes cited answers.
6. An **evidence process** (sealed sets, leakage tests, checksums, tracker) decides what ships, and its honest negatives (verifier NO-GO, rejected levers) are part of the product.

25.2 Consolidated visual: how everything connects



25.3 Reconciliation check (diagrams versus text)

Check	Result
Every component in the consolidated visual appears in the catalogue (Section 7.1)	Yes
Every requirement in Section 4 maps to a component and a verifier (Section 5.2)	Yes
Every ADR corresponds to a built behaviour or is marked Proposed/Open	Yes
Planned items are shown dashed or in the red "PLANNED" node, never as built	Yes
Figures that changed since the 26 Oct status report were re-read from the branch tip: embedding model (bge-m3), point count (2,109), opt-in browser router, advisory, sample capture, golden set	Yes

25.4 Open decisions and what is still unknown

#	Open item	Owner
F3	Operating points per action level; leave shadow mode; default router version (0.4.1 or 0.5.1)	Founder
F5 / D1	Topology and reference devices	Founder
F7	Counsel on SEBI and DPDP questions	Founder + counsel
F4 / N2	Second label reviewer; Hindi reader for the Hindi/Hinglish slice	Founder
F1	1,000+ redacted statement rows from several people	Founder + contributors
F2	Feedback endpoint, consent wording, retention	Founder + counsel
A2	Real workload numbers	Pilot data
A4	Whether `rag_documents` / `rag_feedback` are applied to the live Supabase project	Founder
Hosting	Where the engine runs for people other than the founder	Founder

25.5 Next actions

1. Run the shadow week with the router flags on; read `smlReport()` after about 300 messages.
2. Answer F3 with the false-alarm numbers you accept per level.
3. Name the second reviewer and the Hindi reader; use the prepared review pages.
4. Ask 3-5 people for redacted categories using the contributor guide and validator.
5. Decide topology D1 and engage counsel (F7) before any public exposure.
6. After those inputs, build the action ladder behind a flag, harden the browser note-level router, grow the golden set, and add logs, a dashboard and alerts.